When Is Surrogate-Assisted Multiobjective Ensemble Weighting Trustworthy?

A Replicated Study of Mixture Designs, Scheffé Surfaces and Normal Boundary Intersection on the Classifier-Weight Simplex

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Abstract

The weights of a soft-voting classifier ensemble are non-negative and sum to one, so choosing them is a mixture experiment on a simplex. Building on an established framework combining mixture designs of experiments, response-surface metamodels and Normal Boundary Intersection — already applied to forecast-combination and neural-network ensemble weights — we ask not whether that pipeline yields a Pareto front for classifier ensembles, but when the front it yields can be trusted. The transfer is not routine: the area under the ROC curve is piecewise constant in the weights, the decision variables are themselves the mixture so that anchor points are vertices of the feasible region, and deployment cost is a step function of the support. We therefore decompose the pipeline into three arms differing only in where objectives and anchors come from — surrogate objectives with surrogate anchors, the same surfaces with anchors from real out-of-fold optima, and a metamodel-free arm — and add four controls the framework lacks: external validation of every surface on 100 unseen compositions behind a pre-registered reliability gate, revalidation of every candidate on the real objectives, an empirical Pareto reference of at least 10^5 points built independently of the surrogate, and an untouched holdout. The study is replicated over four public binary tabular datasets $\times$ 30 outer partitions, paired by partition with overlap-corrected inference. Surrogate adequacy is both dataset- and metric-dependent: ROC-AUC surfaces pass the gate in 0, 5, 16 and 30 of 30 partitions, log-loss surfaces in 30, 3, 12 and 30. Anchor construction, not interior surrogate accuracy, is the first-order failure mode — real anchors improve both primary endpoints in every partition of the two datasets whose surfaces fail and in 24 of 30 elsewhere — and the gate detects unusable surfaces but not misplaced anchors, the two actively diverging on one dataset. The classical synergism criterion holds for the leading classifier pair everywhere, yet the real blend beats its better member on only one dataset. Finally, the continuous cost these methods optimize disagrees with the step cost deployment pays about which method wins in up to 24 of 30 partitions. The principal limitation is that four datasets support statements about partition sensitivity within a dataset, but only a four-point pattern across them.

Keywords: ensemble weighting; mixture design of experiments; Scheffé canonical polynomials; response surface methodology; normal boundary intersection; surrogate-assisted multiobjective optimization; inference cost; replicated evaluation.

1 Introduction

Combining several trained classifiers by a weighted average of their predicted probabilities is one of the oldest and most reliable devices in applied machine learning (Hansen and Salamon,

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1990; Wolpert, 1992; Kittler et al., 1998; Caruana et al., 2004). The weights that govern such a combination are not free parameters in a box. For the combined output to remain a probability they must be non-negative and sum to one, a constraint recognized as beneficial in its own right since Breiman (1996) showed that non-negativity, and in practice normalization, is what makes stacked combinations stable. The feasible set is therefore the unit simplex, and choosing ensemble weights is a *compositional* optimization problem.

That observation has a direct methodological consequence. Optimization over a simplex is the subject of a mature branch of experimental design: mixture experiments, whose canonical polynomial models were introduced by Scheffé (1958, 1963) and developed into standard practice by Cornell (2002). The components are the classifiers, a pure vertex is a single model used alone, the centroid is uniform soft voting, and a Scheffé canonical polynomial is the natural response surface, because ordinary polynomials with an intercept are singular under the sum-to-one constraint. This correspondence is exact, and it has already been exploited: mixture designs have been used to weight portfolio components (Mendes et al., 2016; Leal et al., 2022), to weight the members of a time-series forecast combination, where a simplex lattice over the combination weights feeds Scheffé models that are then optimized by Normal Boundary Intersection (Bacci et al., 2019), to weight neural-network ensembles (Moreira et al., 2021), and to weight a three-network load-forecasting ensemble through a simplex-lattice design, a factor-analytic surrogate and NBI (Rocha et al., 2025). A closely related framework combining mixture designs, response-surface metamodels and NBI with a mixture-design post-Pareto stage was developed for multiobjective engineering optimization (Pereira et al., 2025). Most directly, Kwon et al. (2024) treat the weights of five heterogeneous *classifiers* as mixture components, fit a Scheffé polynomial to accuracy or F_1 over the weight simplex, reduce it by backward elimination and maximize it under $\sum_i w_i = 1$ on eleven tabular binary datasets.

We therefore claim no part of that construction. Mixture design over combination weights, a Scheffé surrogate of performance over the simplex, and NBI on that surrogate are established prior work — individually, in combination, and specifically for classifier ensembles. What has not been established is whether the construction remains *trustworthy* once the problem becomes genuinely multiobjective, and that is the question of this paper.

1.1 Why the transfer is not routine

The engineering applications in which this framework matured share three properties that classifier ensembles do not.

First, *the responses are smooth.* Tool life, surface roughness and process cost vary smoothly with cutting speed or feed rate, and a low-order polynomial is a defensible local model. The area under the ROC curve is a rank statistic: it is piecewise constant in the weights, with zero gradient almost everywhere and kinks where the induced ordering of samples changes. No experimental design repairs a misspecified model class, and a smooth surrogate fitted to a step-like response will misrepresent it in a specific, systematic direction.

Second, *the decision variables are the mixture.* When the simplex holds scalarization weights, as in most of the lineage above, the anchors that NBI needs are optima of the surrogate in some other space. When the simplex holds the decision variables themselves, the anchors are points of the feasible region — frequently its vertices — and an anchor error is a displacement of the geometry within which every subproblem is posed. NBI derives its utopia point, its objective normalization and its quasi-normal directions from the payoff matrix of individual optima (Das and Dennis, 1998), so errors there propagate globally rather than locally. The unreliability of ideal and nadir estimates is known in the classical multiobjective literature (Messac et al., 2003; Deb et al., 2010), but it is posed there as a question about estimating the true objectives, not about a surrogate standing in for them.

Third, *one objective is a step function.* Deployment cost is not paid in proportion to a weight: a model that receives any non-negligible weight must be loaded and executed, and one that receives none costs nothing. A multiobjective method that requires differentiable objectives can only ever optimize a continuous relaxation of that cost, and there is no guarantee that the relaxation and the quantity actually paid agree about which solution is best.

Each of these breaks a different part of the pipeline — the model class, the search geometry, and the objective definition — and none of them is visible from the aggregate fit statistics that such studies conventionally report.

1.2 Approach

We evaluate the framework by decomposing it. Because the ensemble output is linear in the weights and the base-model probabilities can be cached, evaluating any weight vector on the true out-of-fold objectives costs one matrix–vector product, between 0.4 and 2.1 milliseconds in our setting. This unusual affordability lets us do something that is normally out of reach in surrogate-assisted optimization: run the *same* NBI three times per replication, changing only where the objectives and the anchors come from.

- **NBI-A** optimizes the Scheffé surfaces with anchors taken from the surfaces' own optima. This is the inherited construction and serves as the control condition.
- **NBI-B** optimizes the same surfaces but takes its anchors from the real single-objective optima. The difference between A and B isolates *anchor misplacement*.
- **NBI-C** discards the metamodel and runs the same NBI on the exact out-of-fold objectives. The difference between B and C isolates the residual effect of *surrogate error in the interior*.

Around this decomposition we place four controls that the lineage does not contain: an external validation of every surface on 100 compositions it has never seen, with a pre-registered reliability gate; revalidation of every candidate produced by any method on the true objectives before any indicator is computed; an empirical Pareto reference of at least 10^5 points per replication, built independently of the surrogate and checked for convergence; and an untouched holdout that never participates in fitting, weighting, thresholding or selection.

The whole pipeline is then replicated. Four public binary tabular datasets are each partitioned 30 times, giving 120 replications with recorded seeds. Because partitions of one dataset overlap heavily, we treat the replication distribution as a measure of partition sensitivity rather than as a sample from a population: all comparisons are paired by partition and tested with the Nadeau–Bengio correction (Nadeau and Bengio, 2003), the dataset is the unit of generalization, and nothing is pooled across the 120 replications.

1.3 Research questions

RQ1 How accurately do Scheffé surfaces predict ensemble performance on compositions they have not seen, and does this depend on the dataset, on the metric, or on both?

RQ2 How stable are the fitted surfaces across resampling, and does the classical mixture-design reading of the interaction coefficients survive confrontation with the real objectives?

RQ3 What goes wrong when NBI is anchored on surrogate optima?

RQ4 Does replacing surrogate anchors with real ones repair the front, and does the reliability gate predict when this is necessary?

RQ5 How does surrogate-assisted NBI compare with metamodel-free NBI, and what does the difference cost?

RQ6 How large is the actual conflict between discrimination and calibration?

RQ7 Does the definition of deployment cost change which method wins?

RQ8 Do solutions selected on out-of-fold data transfer to an untouched holdout?

RQ9 Would a third of the replications have produced the same conclusions?

1.4 Contributions

1. **A controlled decomposition of a surrogate-assisted multiobjective pipeline into surrogate error, anchor misplacement and solver behaviour**, through three NBI variants that differ in exactly two binary choices. To the best of our literature search, no prior study computes NBI anchors both from a surrogate and from the true objectives and compares the resulting fronts. The nearest precedent for the pipeline itself, Kwon et al. (2024), optimizes one accuracy metric at a time and therefore has no anchors, no front and no such decomposition.

2. **Evidence that anchor construction, not surrogate accuracy, is the first-order failure mode.** Real anchors improve both primary endpoints in 30 of 30 partitions on the two datasets whose ROC-AUC surfaces almost never pass the gate, and in 24 of 30 on the other two. Removing the surrogate entirely, once anchors are correct, adds much less. The fix is also the cheap one: real anchors are the single-objective optima a practitioner already computes.

3. **A demonstration that an external reliability gate detects unusable surfaces but not misplaced anchors.** On one dataset the two properties actively diverge: the partitions in which surrogate-anchored NBI collapses are precisely those in which the parsimony rule selects a surface that fits held-out compositions *better* and passes the gate more often. Surface quality and argmax quality are different properties, and only the second governs NBI.

4. **A replicated evaluation architecture for ensemble-weight optimization** — external surrogate validation, real-objective revalidation of every candidate, an empirical Pareto reference independent of the methods under test, and untouched-holdout confirmation, over 4 datasets $\times$ 30 partitions with overlap-corrected paired inference. Three findings that this architecture overturned, including two artifacts that survived an earlier ten-partition analysis, are reported explicitly.

5. **Two findings about how these surfaces should be read.** The classical synergism criterion $\beta_{ij} > |\beta_i - \beta_j|$ is satisfied by the fitted surfaces for the leading pair on all four datasets in every partition, yet the real blend beats its better member on only one; across all pairs the surrogate over-predicts blending gains 20 times and under-predicts none. And the continuous cost that these methods can optimize disagrees with the step cost that deployment pays often enough to change the winning method in up to 24 of 30 partitions.

We deliberately do not claim that NBI outperforms other multiobjective optimizers, that it produces more uniformly spaced fronts — it does not, after revalidation — or that any method here is a general-purpose recommendation. The contribution is a set of conditions under which an established framework can and cannot be trusted in a new problem class, and the controls needed to tell the difference.

2 Related work

2.1 Classifier ensemble weighting

Combining classifiers by a weighted vote or a weighted average of probabilities is long established (Hansen and Salamon, 1990; Kittler et al., 1998; Dietterich, 2000; Kuncheva, 2014). The question of how to set the weights has two classical answers. Stacked generalization (Wolpert, 1992) learns a meta-model on out-of-fold predictions; Breiman (1996) showed that constraining the combination coefficients to be non-negative — and, in his formulation, to sum to one — is what makes stacking stable, an observation extended by LeBlanc and Tibshirani (1996) and formalized in the super-learner literature (van der Laan et al., 2007; LeDell et al., 2016). Greedy forward selection with replacement (Caruana et al., 2004) remains the practical default in automated machine learning. Fumera and Roli (2005) give the theoretical analysis of weighted averaging, and a broad recent literature optimizes ensemble weights numerically, by convex programming, metaheuristics or Bayesian search (Zhang and Zhou, 2011; Yao et al., 2018; Shahhosseini et al., 2022; Hassan et al., 2026; Xu et al., 2026).

Two observations matter for this paper. First, the simplex constraint is standard but is almost always treated as a constraint to be respected rather than as an experimental region to be designed over: we found no study in this cluster that runs a statistical design over the weight simplex and fits a response surface to it. Second, essentially all of this work is single-objective, optimizing one criterion at a time.

2.2 Multiobjective ensemble optimization

Multiobjective formulations of ensemble learning divide by what the decision variables are. A large family evolves the *members* under accuracy-versus-diversity objectives (Abbass, 2003; Chandra and Yao, 2006; Jin and Sendhoff, 2008). A second family selects a *subset*, with binary membership variables, trading error against ensemble size or confidence (Dos Santos et al., 2008; Qian et al., 2015). A third, and the one relevant here, optimizes *continuous combination weights* under conflicting criteria: Ekbal and Saha (2011) and Ekbal and Saha (2013) optimize per-class vote weights with multiobjective simulated annealing (Bandyopadhyay et al., 2008) under recall against precision; Onan et al. (2016) and Rezaei et al. (2019) use multiobjective differential evolution on classifier weights; Zhao et al. (2018) add a sparsity ratio as a third objective; Vrugt et al. (2006) optimize Bayesian model-averaging weights, which lie on a simplex by construction, against several forecast-skill metrics; and Galván et al. (2026) optimize voting weights jointly with feature selection. Surveys organize the field by generation, selection and combination (Ribeiro and Reynoso-Meza, 2020).

Multiobjective optimization of continuous ensemble weights is therefore *not* new, and we do not claim it. What is absent from this cluster is any use of a designed experiment and a fitted response surface over the weight simplex, any use of a classical scalarization method such as NBI, and — with the partial exception of the sparsity term in Zhao et al. (2018) — any treatment of deployment cost as an objective. The evaluation practice is also different: fronts are typically compared between algorithms, not scored against an independently constructed reference, and are rarely replicated over resampled partitions with corrected inference (Demšar, 2006).

2.3 Mixture experiments and response surfaces

Mixture experiments address exactly the constraint set of Equation (2). Scheffé (1958) introduced the simplex-lattice designs and the canonical polynomials that carry his name, followed by the simplex-centroid design (Scheffé, 1963); Cornell (2002) is the standard reference, and Myers et al. (2016) places mixture designs within response-surface methodology. Piepel (1982) is directly relevant to how we read our coefficients: because $\sum_i w_i = 1$, mixture-model coefficients are not component effects in the ordinary sense, and separate measures are needed to interpret them.

The classical reading of a positive interaction coefficient is *synergistic blending*, defined as a departure of the blend above the linear (additive) prediction from the pure components (Cornell, 2002). That is a statement relative to the chord, not relative to the better pure component, and the distinction turns out to matter (Section 5.2).

Mixture designs are beginning to appear in machine learning. Karl et al. (2023) apply self-validated ensemble modelling to mixture designs; Leal et al. (2025) estimate the weights of a Gaussian mixture model through a simplex-lattice design and a response surface on the log-likelihood; and a fast-growing literature treats the proportions of a training-data mixture as a simplex and fits predictive laws over it (Xie et al., 2023; Ye et al., 2025; Liu et al., 2025; Chen et al., 2025; Thudi et al., 2025). These share our formalism but not our object: none models the performance of a weighted *prediction* ensemble.

The exception, and the closest prior art to this study, is Kwon et al. (2024). They treat the weights of five heterogeneous classifiers as mixture components, obtain the generalization metric at each point of a mixture design, fit a Scheffé canonical polynomial with binary interaction terms to accuracy or F_1, reduce it by backward elimination at $\alpha = 0.05$, and maximize the reduced polynomial subject to $\sum_i w_i = 1$ with SLSQP and related solvers, on eleven tabular binary datasets against stacking meta-models. Elements 1–3 of our own pipeline — the simplex formulation, the mixture design over classifier weights and the Scheffé surrogate of ensemble performance — are therefore established prior art for classifiers, not only for forecasts and neural networks, and we claim none of them. What that study does not do is what this paper is about: it optimizes one scalar metric at a time, so it has no Pareto front, no anchors and no NBI; it validates the fitted polynomial only in-sample, with no external validation on unseen compositions and no admissibility criterion; it has no independent reference front and no Pareto indicators; it carries no cost objective; and it reports no replication over resampled partitions and no paired inference.

2.4 The DoE–RSM–NBI framework and its application to ensemble weights

Normal Boundary Intersection (Das and Dennis, 1998) generates well-distributed Pareto points by solving, for each convex weight vector, a subproblem that advances from the convex hull of individual minima along a quasi-normal direction. It is widely combined with response-surface metamodels in engineering optimization, and a substantial body of work applies NBI to polynomial surfaces fitted from designed experiments (Brito et al., 2014; Costa et al., 2016; Lopes et al., 2016; Naves et al., 2017; Belinato et al., 2019; de Almeida et al., 2022).

Within that tradition, mixture designs have been used to schedule the NBI weights and to model post-Pareto criteria over the weight simplex. Mendes et al. (2016) and Leal et al. (2022) model portfolio proportions with mixture designs; Bacci et al. (2019) builds a simplex lattice over time-series forecast-combination weights, fits Scheffé models to factor scores of residual accuracy metrics and applies NBI to those models; de Paula et al. (2019) crosses a mixture design over objective weights with genetic-algorithm hyperparameters. Pereira et al. (2025), of which the present first author is a co-author, extends the framework with a mixture-design post-Pareto stage in which Scheffé polynomials of a generalized distance and of the entropy of the weights are themselves optimized by a second NBI. Its anchors, like those of the rest of this literature, are the individual optima of the surrogate surfaces — the configuration we label NBI-A. The first author's master's dissertation (Ribeiro, 2026) applies the same framework to hyperparameter optimization.

Most directly, Moreira et al. (2021) determine neural-network ensemble weights by a mixture design, and Rocha et al. (2025) combine a $\{3, 5\}$ simplex-lattice design over the mixing weights of three neural networks, a factor-analytic surrogate of correlated error metrics, NBI over those surrogates, and an entropy-based post-Pareto choice, for short-term load forecasting. That is the present pipeline applied to a regression ensemble, and its stated first contribution is casting ensemble-weight definition as a structured mixture-design problem.

We therefore claim no part of this construction as new. Our contribution begins where this literature stops: none of these studies validates its mixture surrogate on unseen compositions, computes anchors from anything other than the surrogate, revalidates its Pareto candidates on the true objectives, scores them against an independently built reference, or replicates over resampled partitions.

2.5 Surrogate-assisted multiobjective optimization

The broader surrogate-assisted literature is well surveyed (Jin, 2011; Tabatabaei et al., 2015; Chugh et al., 2019; Deb et al., 2019), with prominent methods including ParEGO (Knowles, 2006), MOEA/D-EGO (Zhang et al., 2010) and SMS-EGO (Ponweiser et al., 2008). This literature is acutely aware that a surrogate may mislead, and addresses it primarily through *model management*: deciding which points to evaluate expensively so that the surrogate improves where it matters. The reliability of the extreme points themselves has been studied for the true objectives — the difficulty of estimating the nadir point is a recognized problem (Deb et al., 2010), normalization strategies are known to affect performance (He et al., 2021), and the normalized normal constraint was introduced partly to make anchor-based normalization better behaved (Messac et al., 2003).

What we did not find, in this cluster or any other, is a study that computes the anchors of a CHIM-based method both from a surrogate and from the true objectives and compares the resulting fronts, or one that revalidates a complete surrogate-generated Pareto set on the true objectives and scores it with IGD$^+$ and hypervolume against an *empirically constructed* reference. The closest external analogue we located is Abolghasemian et al. (2022), which fits regression metamodels from a designed experiment, validates the metamodels by fit statistics, and runs a modified NBI on them without re-evaluating the resulting Pareto solutions on the simulation — precisely the NBI-A pattern this study is built to expose.

2.6 Cost-aware ensemble deployment

Prediction-time cost has been an explicit objective since Turney (2000). The literature prices it in three ways: as the cost of a fixed *deployed set*, as in ensemble pruning and hardware-aware or energy-aware ensemble selection (Zhou et al., 2002; Ji and Li, 2023; Maier et al., 2024; Gunasekaran et al., 2022; Moreira et al., 2026); as *per-example adaptive execution* through cascades and early exits (Xu et al., 2013; Trapeznikov et al., 2013; Nan and Saligrama, 2017; Chen et al., 2020); and as a *continuous penalty* inside a scalarized training objective (Xu et al., 2012; Zhang and Zhou, 2011). Automated machine-learning systems increasingly expose inference-time constraints during ensemble construction (Erickson et al., 2020; Feurer et al., 2022; Borchert et al., 2022).

The distinction we need falls between these categories. Our weights induce a support, so deployment pays a step cost, while the optimizers can only minimize its continuous relaxation. To the best of our literature search, no prior work contrasts these two cost functions for the same weighted ensemble and reports that they disagree about which method wins. Where a related question has been settled it went the other way: for budgeted learning with feature re-use, the support-cost programme is provably solved by its linear relaxation (Nan et al., 2015). Our setting does not inherit that guarantee, and Section 5.6 shows the two definitions disagreeing in up to 24 of 30 partitions.

2.7 The gap

Ensemble weights on a simplex, mixture designs over them, and Scheffé surrogates of ensemble performance all exist, and for classifier ensembles specifically they exist together in Kwon et al.

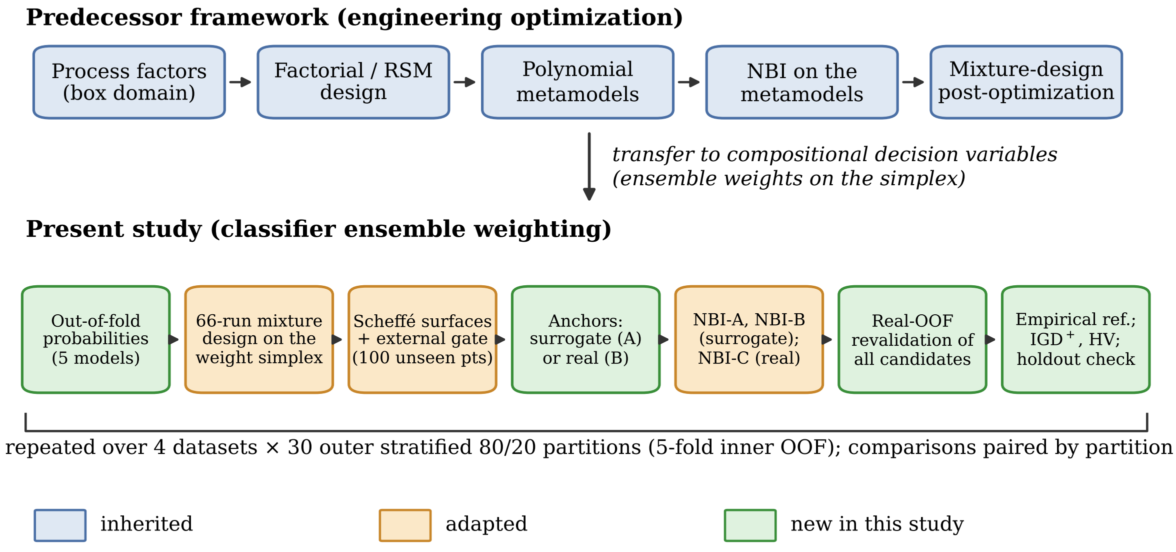

Figure 1: The predecessor DoE–RSM–NBI framework (top) and its transfer to classifier ensemble weighting (bottom). Blue: inherited. Orange: adapted because the decision variables are compositional and the responses are machine-learning metrics. Green: introduced in this study. The entire lower lane is repeated on 30 outer partitions of each of four datasets, and every comparison is paired by partition.

(2024); adding NBI on those surrogates exists in Rocha et al. (2025) for a regression ensemble and in Bacci et al. (2019) for forecast combination. Multiobjective optimization of continuous ensemble weights exists. Cost-aware ensemble selection exists. What does not exist, to the best of our literature search, is an evaluation of whether this pipeline can be trusted once it is both multiobjective and applied to classifiers: no external validation of the mixture surrogate on unseen compositions, no comparison of surrogate-derived against real anchors, no metamodel-free control arm, no revalidation of candidates on the true objectives against an independent empirical reference, no replicated partition study of the whole pipeline, and no contrast between the cost that is optimized and the cost that is paid.

The gap is therefore not a missing method. It is the absence of the controls that would tell a practitioner when the existing method's output can be believed — which is what this paper supplies.

3 Methodology

This section states the optimization problem, the mixture-design and response-surface machinery inherited from the predecessor framework, and the three Normal Boundary Intersection (NBI) variants that separate the effect of surrogate fidelity from the effect of anchor construction. Figure 1 shows the pipeline and marks each element as inherited from the predecessor framework, adapted for compositional decision variables, or new in this study.

3.1 The ensemble-weighting problem is a mixture problem

Let M probabilistic base classifiers produce, for a sample x, probability estimates $p_1(x), \ldots, p_M(x)$ in $[0, 1]$. A weighted soft-voting ensemble predicts

$$p_{\mathbf{w}}(x) = \sum_{i=1}^{M} w_i \, p_i(x) = \mathbf{P}\mathbf{w}, \tag{1}$$

where $\mathbf{P} \in [0,1]^{n\times M}$ collects the base-model probabilities for n samples. For $p_{\mathbf{w}}(x)$ to remain a probability for every input, the weights must satisfy

$$w_i \geq 0 \quad (i = 1, \ldots, M), \qquad \sum_{i=1}^{M} w_i = 1, \tag{2}$$

that is, $\mathbf{w} \in \Delta^{M-1}$, the unit simplex in $\mathbb{R}^M$. Equation (2) is the canonical constraint set of a *mixture experiment*: the components are the classifiers, the pure vertex $w_i = 1$ is the single model i used alone, and the centroid is uniform soft voting. The experimental region is a simplex, not a box, so factorial designs and ordinary polynomials with an intercept are inapplicable — the sum-to-one constraint makes the design matrix singular — and the appropriate formalism is the mixture design with a Scheffé canonical polynomial. This correspondence, rather than any performance argument, is why the framework transfers at all.

Three objectives are considered, all functions of $\mathbf{w}$ through Equation (1):

$$f_1(\mathbf{w}) = -\,\mathrm{AUC}(y, \mathbf{P}\mathbf{w}) \qquad \text{(discrimination, maximized)}, \tag{3}$$

$$f_2(\mathbf{w}) = \frac{1}{n}\sum_{k=1}^{n}\Big[-\,y_k \log p_{\mathbf{w}}(x_k) - (1-y_k)\log\big(1 - p_{\mathbf{w}}(x_k)\big)\Big] \qquad \text{(calibration, minimized)}, \tag{4}$$

$$f_3(\mathbf{w}) = c(\mathbf{w}) \qquad \text{(inference cost, minimized)}. \tag{5}$$

The three differ qualitatively, and that heterogeneity is the point of the study. The log-loss f_2 is convex and smooth in $\mathbf{w}$. The area under the ROC curve is a rank statistic: f_1 is piecewise constant in $\mathbf{w}$, with gradients that are zero almost everywhere and undefined on the boundaries between orderings, so it is not merely non-convex but non-smooth in a way that defeats gradient-based certification. The cost f_3 is discussed next.

3.2 Two cost models: a relaxation and a deployment cost

Let c_i be the measured inference cost of model i in milliseconds per 1,000 predictions, including its preprocessing. Deployment cost is a step function of the weight vector: a model that receives *any* non-negligible weight must be loaded, fed and executed, and one that receives none costs nothing. This gives the *support cost*

$$c_{\sup}(\mathbf{w}) \;=\; \sum_{i=1}^{M} c_i\, \mathbf{1}[w_i > \varepsilon]\,, \qquad \varepsilon = 10^{-3}, \tag{6}$$

which is discontinuous and therefore unusable by a gradient-based method. Every optimizer here instead sees the continuous relaxation

$$c_{\mathrm{w}}(\mathbf{w}) \;=\; \sum_{i=1}^{M} w_i c_i, \tag{7}$$

the natural linear mixture response, which coincides with $c_{\sup}$ at the vertices but not on the interior: spreading a little mass over an expensive model is cheap under Equation (7) and fully priced under Equation (6). We therefore optimize Equation (7) and additionally score the *same* solutions under Equation (6), reporting every comparison under both. Section 5.6 shows that the choice changes which method wins.

3.3 Mixture design and Scheffé surrogates

Design. With $M = 5$ the design is a fixed 66-run mixture design: the union of the $\{5, 3\}$ and $\{5, 2\}$ simplex lattices, the overall centroid, the 5 axial points, the 5 quaternary centroids and the 10 midpoints between the overall centroid and the ternary centroids. It therefore places design points on the 5 pure vertices, all 10 binary edges, the ternary and quaternary faces and uniform voting, giving the surrogate support on every face a sparse ensemble could occupy.

Models. For a response y the Scheffé canonical polynomials of order 1, 2 and the special cubic are

$$\text{linear:} \quad \hat{y} = \sum_i \beta_i w_i, \tag{8}$$

$$\text{quadratic:} \quad \hat{y} = \sum_i \beta_i w_i + \sum_{i<j} \beta_{ij} w_i w_j, \tag{9}$$

$$\text{special cubic:} \quad \hat{y} = \sum_i \beta_i w_i + \sum_{i<j} \beta_{ij} w_i w_j + \sum_{i<j<k} \beta_{ijk} w_i w_j w_k, \tag{10}$$

fitted by ordinary least squares without an intercept and without pure quadratic terms, since $\sum_i w_i = 1$ makes them redundant. No term elimination is performed: the canonical form is kept intact so that β_i is the predicted response at vertex i and β_{ij} measures the departure of the i–j edge from linear blending.

External validation and the reliability gate. Where the lineage reports in-sample fit statistics, each replication here additionally evaluates 100 compositions drawn from Dirichlet($\mathbf{1}$) (60) and Dirichlet($0.5 \cdot \mathbf{1}$) (40) that are not in the design and never used for fitting, giving an external R^2_{ext} and Spearman ρ_{ext}. *Order selection* then uses a parsimony rule — the lowest order whose external RMSE is within 10% of the best — so it is driven by out-of-design accuracy rather than in-sample fit. *Admissibility* uses a pre-registered *reliability gate*: a surface is reliable only if

$$R^2_{\text{ext}} \geq 0.5 \quad \text{and} \quad \rho_{\text{ext}} \geq 0.9. \tag{11}$$

The thresholds were fixed before the replicated study and were never adjusted. The gate is a hypothesis under test in this work, not a filter applied to the results: surrogate-based optimization is run and reported *regardless* of the gate outcome, precisely so that the gate's predictive value can be measured (Section 5.1).

3.4 Normal Boundary Intersection on the simplex

NBI (Das and Dennis, 1998) generates an even spread of Pareto points by intersecting the boundary of the attainable set with normals to the convex hull of individual minima. For m objectives $F = (F_1, \dots, F_m)$ let $\mathbf{w}^{*(k)}$ minimize F_k; the payoff matrix Φ has columns $F(\mathbf{w}^{*(k)})$, its diagonal is the utopia point F^{u}, and objectives are normalized by the utopia and pseudo-nadir points extracted from Φ. Writing $\bar{\Phi}$ for the normalized payoff matrix and $\hat{n} = -\bar{\Phi}\mathbf{1}$ for the quasi-normal direction, each β on the unit simplex of *scalarization* weights defines the subproblem

$$\max_{\mathbf{w},\, t} \; t \quad \text{s.t.} \quad \bar{\Phi}\beta + t\,\hat{n} = \bar{F}(\mathbf{w}), \quad \mathbf{w} \in \Delta^{M-1}. \tag{12}$$

Two implementation points matter. The simplex constraint is eliminated by optimizing over $M-1$ free variables with $w_M = 1 - \sum_{i<M} w_i$, with a projection step returning the nearest feasible composition when an iterate leaves the simplex; without it, subproblems near a vertex can terminate infeasible. And for vertex values of β the solution is the corresponding anchor, returned directly rather than re-derived. We solve 66 subproblems per replication on the same β lattice for all variants, so the arms are compared at identical points of the CHIM.

The three variants. The variants differ in exactly two binary choices — where the objectives come from, and where the anchors come from — and in nothing else:

Variant	Objectives in Equation (12)	Anchors / payoff matrix	Real objective evaluations
NBI-A	Scheffé surfaces	optima of the Scheffé surfaces	0
NBI-B	Scheffé surfaces	real out-of-fold optima	0 (anchors reused)
NBI-C	real out-of-fold objectives	real out-of-fold optima	$\approx 4.2 \times 10^5$

NBI-A is the construction used throughout the lineage: everything, including the payoff matrix, comes from the metamodel. NBI-B changes one thing — the anchors are the real optima of Section 3.5 — so A versus B isolates *anchor misplacement* with the surfaces held fixed. NBI-C removes the metamodel entirely, so B versus C isolates the residual effect of *surrogate error in the interior* once anchors are correct. Because f_1 is piecewise constant, NBI-C uses finite-difference steps of 10^{-3}, multistart on the simplex, and accepts an iterate whose equality residual is below 10^{-3} even without certified optimality, whereas NBI-A and NBI-B optimize smooth polynomials and require full SLSQP certification. This asymmetry follows from the objective, not from tuning, and is reported with the results.

3.5 References, comparators and revalidation

Single-objective references. Each replication computes the best single model, uniform voting, logistic stacking, the SLSQP log-loss optimum (convex, hence global), a derivative-free ROC-AUC optimum over 4×10^4 sampled compositions with local refinement, and the optima of the two Scheffé surfaces. The log-loss and AUC optima double as the real anchors of NBI-B and NBI-C. SLSQP on the convex log-loss remains the correct tool for the single-objective problem; NBI is not proposed as a replacement for it.

Comparators. Two cheap alternatives are run on every replication: random weighted scalarization, which draws 66 scalarization weight vectors from Dirichlet($\mathbf{1}$) and minimizes the corresponding weighted sum of the *same* Scheffé surfaces with the same real anchors used by NBI-B; and a 66-point random Dirichlet($\mathbf{1}$) sample of the weight simplex. The second is matched to the NBI sets only in the *number of returned candidates* and is not compute-matched; it functions as a floor check on the geometry of the reference front, not as an efficiency baseline, and is reported as such throughout.

Revalidation. Every candidate from every method — both NBI variants on surrogates, NBI-C, both comparators, the 66 design runs and the references — is re-evaluated on the exact out-of-fold objectives before any indicator is computed, with non-dominated filtering applied to those revalidated values. No result in this paper is read off a surrogate.

3.6 Empirical Pareto reference and quality indicators

Comparing approximation sets requires a reference, and the true Pareto set of Equation (1) over Δ^{M-1} is unknown. Each replication therefore constructs an *empirical Pareto reference*: at least 10^5 compositions from Dirichlet($\mathbf{1}$) and Dirichlet($0.3 \cdot \mathbf{1}$) in equal parts, augmented with the simplex lattice, all vertices and edges, and a 40-cap ε-constraint sweep, evaluated on the real objectives and filtered to the non-dominated set. An independent sample of 2×10^4 points checks convergence: if it displaces more than 5% of the front, the reference is enlarged and rechecked, up to three rounds. The final reference is the non-dominated union of this sample with every candidate set of the replication, making it at least as good as any method scored against it.

We call this an empirical reference, never a true Pareto front: it is a sample-based upper bound, and because candidate sets contribute to it, a method contributing many points is partly graded against itself.

Approximation sets are scored on min–max normalized objectives by generational distance, inverted generational distance, IGD$^+$ (Ishibuchi et al., 2015), Schott spacing, the exact hypervolume with reference point 1.1 on each axis, expressed as a ratio to the reference hypervolume, coverage, and the joint non-dominated fraction. IGD$^+$ and the hypervolume ratio are pre-registered as the two primary endpoints; the others are descriptive.

Table 1: Datasets. Rows used, number of raw features, positive-class prevalence and training/holdout sizes of each outer partition (stratified 80/20). Kaggle test labels are never used; all selection uses 5-fold out-of-fold predictions within the training part. †200,000-row stratified subsample of the 595,212 available rows, drawn once with a recorded seed.

Dataset	Source	Rows	Features	Prevalence	Train / holdout rows
Santander	Santander Customer Transaction (Kaggle 2019)	200,000	200	0.100	160,000 / 40,000
BNP Paribas	BNP Paribas Cardif Claims (Kaggle 2016)	114,321	131	0.761	91,456 / 22,865
Porto Seguro	Porto Seguro Safe Driver (Kaggle 2017)†	200,000	57	0.036	160,000 / 40,000
UCI credit	Default of credit card clients (UCI 350)	30,000	23	0.221	24,000 / 6,000

4 Experimental protocol

The protocol was fixed before the replicated study and was not changed after any result was observed. It is recorded in a machine-readable manifest with the seed of every replication, and the extension from 10 to 30 replications reused it without modification; Section 5.8 reports what that extension changed.

4.1 Datasets

Four public binary tabular classification problems were selected to span prevalence, dimensionality and feature type (Table 1). Santander is high-dimensional, anonymized and numeric with 10% positives; BNP Paribas mixes numeric and high-cardinality categorical features and is 76% positive; Porto Seguro is strongly imbalanced at 3.6% positives and was subsampled to 200,000 rows by a single stratified draw with a recorded seed, to bound runtime; the UCI credit-default data are small and low-dimensional. For the three Kaggle competition datasets the competition test labels are not public and were never used: all training, weighting, thresholding and model selection use only the published training file, and every performance figure in this paper is computed on data derived from it.

4.2 Base models and leakage control

The ensemble contains $M = 5$ deliberately heterogeneous learners with fixed hyperparameters: ℓ_2-regularized logistic regression; Gaussian naive Bayes; k-nearest neighbours with $k = 100$; a random forest with 200 trees and a minimum of 20 samples per leaf; and gradient-boosted trees (XGBoost, 400 rounds, depth 6, learning rate 0.1, subsample and column subsample 0.8). No hyperparameter search is performed. The zoo is intended to span the accuracy–cost– calibration space, and in particular to include a member that is strong in ranking but poorly calibrated (Gaussian naive Bayes) and a member that is accurate but two orders of magnitude more expensive at inference time (k-nearest neighbours); it is not intended to be a competitive stack.

Preprocessing is fitted inside each training fold and never on validation data: median imputation with missingness indicators for numeric features; standardization for the two distance- and scale-sensitive learners; one-hot encoding with a 0.5% minimum category frequency for the three learners that require it; and ordinal encoding for the two tree ensembles.

Each replication proceeds as follows. The dataset is split by a stratified 80/20 partition into a training part and an untouched holdout. Within the training part, stratified 5-fold cross-validation produces out-of-fold probability estimates $\mathbf{P}$ for all five models; these are the only quantities the design, the surfaces, the optimizers and the selection rules ever see. Each model is then refitted once on the whole training part to produce holdout probabilities $\mathbf{Q}$, which are used exclusively for the confirmation analysis of Section 5.7. Because Equation (1) is linear in $\mathbf{w}$, evaluating any weight vector costs one matrix–vector product, which is what makes an empirical reference of 10^5 points and a metamodel-free NBI arm affordable at all.

Inference cost c_i is measured, not assumed: for each model and replication it is the median of five timed `predict_proba` calls on a fixed 10,000-row batch, including preprocessing, expressed in milliseconds per 1,000 rows.

4.3 Replication, seeds and the statistical unit

The entire pipeline — partition, out-of-fold estimation, design, surfaces, references, three NBI variants, comparators, empirical reference and scoring — is executed $R = 30$ times per dataset, giving 120 replications, with the outer partition, the inner folds and every stochastic component seeded by $20260904 + r$ for replication r. All seeds are persisted in the manifest and the runner is checkpointed by dataset $\times$ replication $\times$ stage, so any stage can be recomputed exactly.

The statistical unit is **one outer partition of one dataset**. Because all methods are evaluated on the same partition, every comparison is *paired*. Two consequences constrain the analysis throughout.

First, the 30 partitions of a dataset overlap heavily: any two training parts share on average 75% of their rows. The replication distribution therefore measures *sensitivity to data partitioning*, not sampling from a population, and the effective information is far below 30 independent runs. Second, and consequently, the dataset is the unit of generalization: results are summarized per dataset and the four dataset-level results are then compared. Nothing is pooled across the 4×30 replications, and no statement of the form "$N = 120$ independent experiments" is made anywhere in this paper.

4.4 Statistical analysis

The primary comparisons are pre-specified: NBI-B versus NBI-A, NBI-C versus NBI-B, NBI-C versus random scalarization, and NBI-C versus the random Dirichlet sample. The primary endpoints are IGD$^+$ and the hypervolume ratio, computed on revalidated objectives against the empirical reference of the same replication, under the weighted cost that the optimizers actually minimized; the support-cost repetition is a pre-specified sensitivity analysis. All other indicators are descriptive.

Paired differences are signed so that $\Delta > 0$ always favours the second-named method: $\Delta\text{HV} = \text{HV}_{\text{new}} - \text{HV}_{\text{ref}}$ and $\Delta\text{IGD}^+ = \text{IGD}^+_{\text{ref}} - \text{IGD}^+_{\text{new}}$. For each dataset $\times$ comparison $\times$ endpoint we report the mean and median with percentile-bootstrap 95% intervals (10,000 resamples, fixed seed), the quartiles and range, the win/tie/loss counts with a tie tolerance of 10^{-4}, the win fraction with a Wilson 95% interval, and the matched-pairs rank-biserial correlation as a distribution-free effect size.

Formal testing acknowledges the overlap between partitions. An ordinary paired t-test on resampled partitions is anticonservative, so the primary test is the Nadeau–Bengio corrected resampled t-test (Nadeau and Bengio, 2003): with $J = 30$ paired differences d_j,

$$t = \frac{\bar{d}}{\sqrt{\operatorname{var}(d)\left(\frac{1}{J} + \rho\right)}}, \qquad \rho = \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{0.20}{0.80} = 0.25, \tag{13}$$

on $J - 1$ degrees of freedom. The correction inflates the variance by a factor of $(1/30 + 0.25)/(1/30) \approx 8.5$, that is, it multiplies standard errors by about 2.9 relative to the naive test. Within each dataset the eight primary tests form one family and receive a Holm correction; no correction is applied across datasets, because datasets are not pooled. A Wilcoxon signed-rank test is reported alongside as a distribution-free check, but it is not corrected for overlap and cannot rescue a non-significant corrected result.

Where a paired-difference distribution is visibly bimodal or heavy-tailed, the mean-based test is uninformative by construction and we say so, reporting the median with its bootstrap interval, the win fraction with its Wilson interval and the rank-biserial correlation as the evidence instead.

Table 2: Surrogate reliability over 30 partitions. Selected Scheffé order (parsimony rule: lowest order within 10% of the best external RMSE), reliability-gate passes ($R^2_{\text{ext}} \geq 0.5$ and Spearman $\rho \geq 0.9$ on 100 unseen compositions) with Wilson 95% intervals, and the median external R^2 with its bootstrap interval. Source: `statistics/reliability_gate_r30.csv`.

Dataset	Response	Selected order (count)	Pass	Wilson 95%	Median R^2_{ext} [95% CI]	Median ρ
Santander	ROC-AUC	quadratic 26, linear 4	0/30	[0.00, 0.11]	-0.333 [-0.559, -0.199]	0.830
Santander	log-loss	quadratic 29, sp. cubic 1	30/30	[0.89, 1.00]	0.971 [0.967, 0.973]	0.994
BNP Paribas	ROC-AUC	linear 22, quadratic 7, sp. cubic 1	5/30	[0.07, 0.34]	0.118 [-0.034, 0.244]	0.908
BNP Paribas	log-loss	linear 21, quadratic 7, sp. cubic 2	3/30	[0.03, 0.26]	-0.180 [-0.246, 0.042]	0.955
Porto Seguro	ROC-AUC	quadratic 26, sp. cubic 2, linear 2	16/30	[0.36, 0.70]	0.519 [0.482, 0.581]	0.924
Porto Seguro	log-loss	linear 20, quadratic 8, sp. cubic 2	12/30	[0.25, 0.58]	0.506 [0.492, 0.580]	0.995
UCI credit	ROC-AUC	quadratic 20, sp. cubic 10	30/30	[0.89, 1.00]	0.992 [0.989, 0.992]	0.995
UCI credit	log-loss	quadratic 27, sp. cubic 3	30/30	[0.89, 1.00]	0.993 [0.991, 0.993]	0.993

Throughout, effect consistency takes precedence over p-values. Frequencies over replications — gate passes, win fractions, sign stability, agreement rates — are reported with Wilson and Jeffreys 95% intervals.

5 Results

Results are organized by research question. All figures and tables derive from the frozen artifacts of the 120 replications, and per-claim provenance is given in the supplementary claims map. Unless stated otherwise, indicators use the weighted cost of Equation (7), which is what the optimizers minimized, and $\Delta > 0$ favours the second-named method.

One fact frames the rest: weighting helps, but modestly. The log-loss optimum improves out-of-fold ROC-AUC over the best single model by 0.0037, 0.0056, 0.0060 and 0.0007 on the four datasets. These gains are what the whole apparatus competes for, and they exceed the differences between the multiobjective methods studied here by one to two orders of magnitude. This paper is about whether the front is trustworthy, not about winning a leaderboard.

5.1 RQ1: How accurately do Scheffé surfaces model ensemble performance?

Surrogate adequacy is neither uniformly good nor uniformly bad; it depends on both the dataset and the response (Table 2, Figure 2).

The ROC-AUC surface passes the gate in 0/30 partitions on Santander, 5/30 on BNP Paribas, 16/30 on Porto Seguro and 30/30 on UCI credit; the log-loss surface passes in 30/30, 3/30, 12/30 and 30/30. Santander is the sharpest case: on the same data and partition the log-loss surface is essentially exact (median $R^2_{\text{ext}} = 0.971$) while the ROC-AUC surface has *negative* external R^2_{ext} (median -0.333) — worse than predicting the mean — even though its rank correlation is a respectable 0.830. Porto Seguro sits on the threshold for both responses, and 30 partitions do not resolve it either way; that is a property of the problem, not a sample-size deficiency.

The failures are misspecification, not extrapolation. The excess of predicted over observed range on the validation points is approximately zero everywhere, so the surrogate is not predicting outside its fitted region; it is a low-order polynomial applied to a rank statistic with plateaus and kinks in the interior of the simplex. As a control, the Brier score — genuinely quadratic in $\mathbf{w}$ — is reproduced with $R^2_{\text{ext}} = 1.000$ in all 120 replications, confirming that the design and fitting procedure are sound and the failure is specific to the response. Figure 3 shows one panel per regime. Fitting a response surface over the weight simplex is therefore not a safe default for classifier ensembles: whether it works must be tested per dataset and per metric, on compositions the surface has not seen.

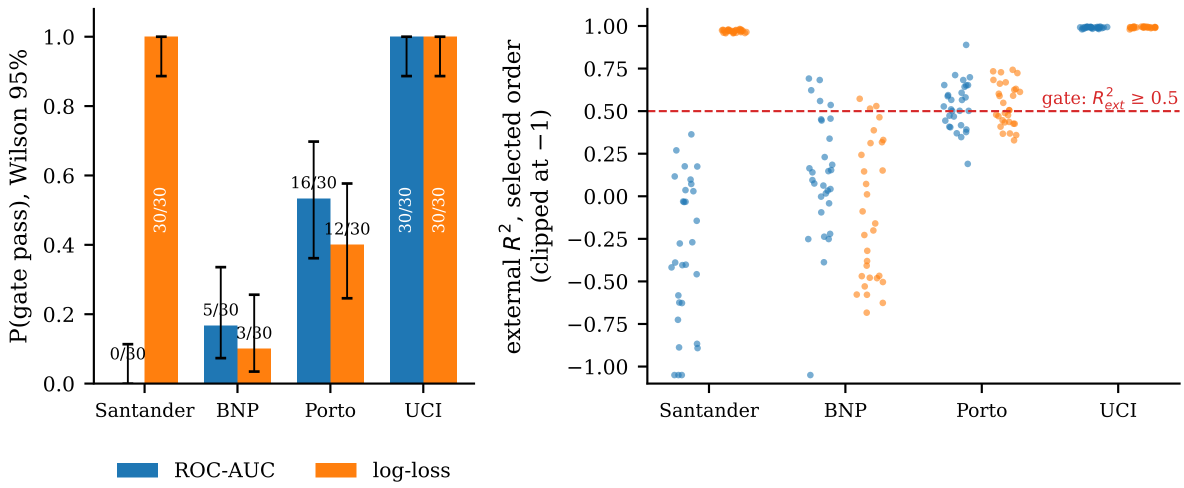

Figure 2: Left: reliability-gate pass frequency per dataset and response over 30 partitions, with Wilson 95% intervals. Right: the external R^2_{ext} of the selected order in each partition; the dashed line is the gate threshold. The four datasets occupy four distinct regimes, and within Santander the two responses fall on opposite sides of the gate.

5.2 RQ2: How stable are the surfaces across resampling, and what do the coefficients mean?

Whatever the surfaces predict, they predict it consistently. Across 30 partitions the sign of each of the four largest interaction coefficients is constant in all eight dataset–response cells; the largest-magnitude interaction is the same term at $R = 10$ and $R = 30$ in all eight, and is the per-partition largest in 30/30 partitions in seven of them (on Porto Seguro's ROC-AUC surface two k-nearest-neighbour pairs alternate, 19/30 against 11/30). Between-partition coefficients of variation for the leading terms lie between 0.011 and 0.17.

The interpretation of those coefficients is the substantive finding, and stating it precisely requires first stating the classical criterion correctly. In mixture experiments a positive β_{ij} denotes *synergistic blending*: a departure of the i–j edge above the chord joining the two pure components (Scheffé, 1958; Cornell, 2002). It is defined against *linear blending*, not against the better component, and the two are not the same question. On the edge $w_i = t$, $w_j = 1 - t$ the quadratic model of Equation (9) gives $\hat{y}(t) = \beta_j + (\beta_i - \beta_j)t + \beta_{ij}t(1-t)$, whose stationary point $t^* = (\beta_i - \beta_j + \beta_{ij})/(2\beta_{ij})$ lies strictly inside $(0, 1)$, with $\hat{y}(t^*) = \beta_j + (\beta_i - \beta_j + \beta_{ij})^2/(4\beta_{ij}) > \max(\beta_i, \beta_j)$, if and only if

$$\beta_{ij} > |\beta_i - \beta_j|. \tag{14}$$

Equation (14) is the criterion a practitioner should apply, and it already guards against the naive reading: a large β_{ij} on an edge joining two very unequal vertices need not produce a blend that beats the better one.

Our finding is not that this criterion is unknown, but that *the fitted surfaces pass it and are still wrong*. For the largest-β pair of every dataset, Equation (14) is satisfied in all thirty partitions (Table 3): the fitted quadratic predicts a genuinely superior interior blend on Santander, BNP Paribas, Porto Seguro and UCI credit alike. Evaluated on the real out-of-fold objectives, that prediction holds on exactly one of them. The real 50/50 blend of the top-β pair is *worse* than its better member in 30/30 partitions on Santander (−0.0050 ROC-AUC), BNP Paribas (−0.0180) and UCI credit (−0.0201); only on Porto Seguro does it beat both members (+0.0098, 30/30). The same pattern holds on the untouched holdout (0/30, 0/30, 29/30, 0/30), and optimally weighting the pair rather than blending it 50/50 does not rescue it: the best attainable gain is +0.0004 on BNP Paribas and +0.0002 on UCI credit, against +0.010 on Porto Seguro.

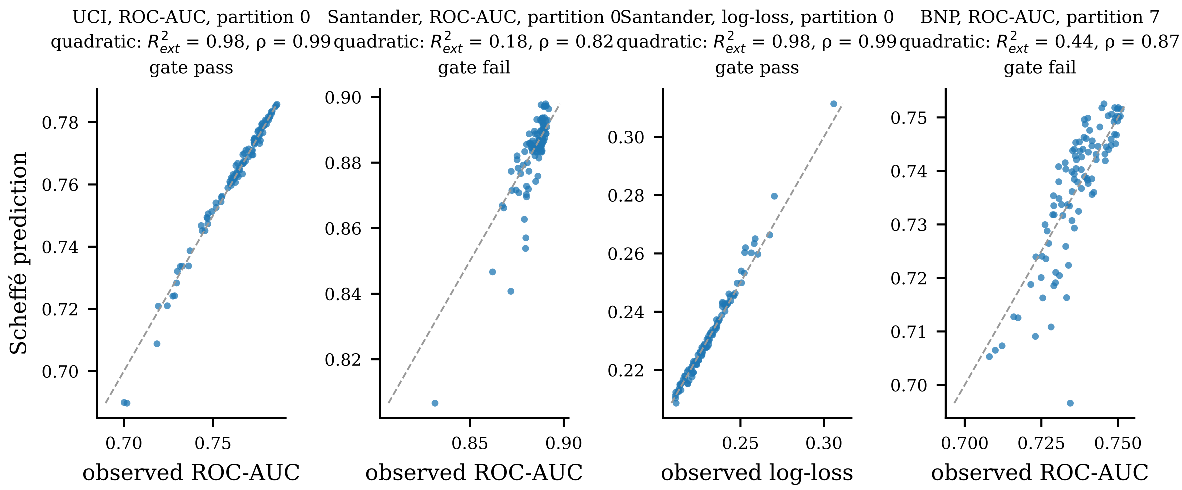

Figure 3: Predicted versus observed response on the 100 unseen Dirichlet compositions of a single partition, for four representative dataset–response pairs spanning the regimes of Table 2. The dashed line is the identity.

Table 3: The classical synergism criterion applied correctly, and what the real objectives say. On the binary edge i–j of a quadratic Scheffé model the fitted surface has an interior optimum exceeding both pure components if and only if $\hat{\beta}_{ij} > |\hat{\beta}_i - \hat{\beta}_j|$. Columns 3–5 evaluate that criterion for the largest-$\hat{\beta}$ pair of each dataset; column 6 gives the partitions in which the *real* out-of-fold 50/50 blend of the same pair beats its better member. The last three columns compare criterion and reality across all ten pairs: agreements, cases where the surface predicts a superior blend that does not exist, and the reverse. The error is systematic and one-directional. Source: `tables/edge_condition_summary.csv`.

		Largest-$\hat{\beta}$ pair, fitted surface			Real blend	All 10 pairs		
Dataset	Pair	$\hat{\beta}_{ij}$	$\lvert\hat{\beta}_i - \hat{\beta}_j\rvert$	criterion met	beats better member	agree	surface over-predicts	under-predicts
Santander	GNB·kNN	0.243	0.130	30/30	0/30	6/10	4	0
BNP Paribas	GNB·XGB	0.148	0.110	30/30	0/30	4/10	6	0
Porto Seguro	GNB·kNN	0.071	0.025	30/30	30/30	7/10	3	0
UCI credit	GNB·XGB	0.130	0.100	30/30	0/30	3/10	7	0

Across all ten pairs the discrepancy is systematic and one-directional. Criterion and reality agree on 6, 4, 7 and 3 of the ten pairs; in every one of the 20 disagreements the surface predicts a superior blend that the real objectives do not deliver, and in none of them does it miss a real one. The surrogate over-predicts blending gains and never under-predicts them.

The mechanism is visible in Figure 4. The largest interaction consistently attaches to the *weakest* vertex — k-nearest neighbours on Santander and Porto Seguro, Gaussian naive Bayes on BNP Paribas and UCI credit — and β_{ij} tracks the vertex-quality gap of the pair (0.163, 0.145, 0.022 and 0.107 for the top pair). A quadratic fitted to a rank statistic on an edge with one poor endpoint absorbs the curvature of that edge into β_{ij} and overshoots its maximum. Consistently, the pairs whose real blend does beat its better member in at least 80% of partitions are different, smaller-β pairs on every dataset except Porto Seguro, and the top-β pair is jointly active on the reference front in only 9% (Santander) and 3% (Porto Seguro) of front points, against 51% and 58% on BNP Paribas and UCI credit.

The practical consequence is narrower and firmer than "β_{ij} does not mean synergy". The classical criterion is correct and should be used in the form of Equation (14) rather than by reading the sign of β_{ij}; but on these ensembles it inherits the misspecification of the surface it is computed from, and it must therefore be confirmed on the real objective before any pair is described as complementary. That confirmation costs one matrix product per pair.

The log-loss interactions are uniformly negative and their sign carries no information: log-loss

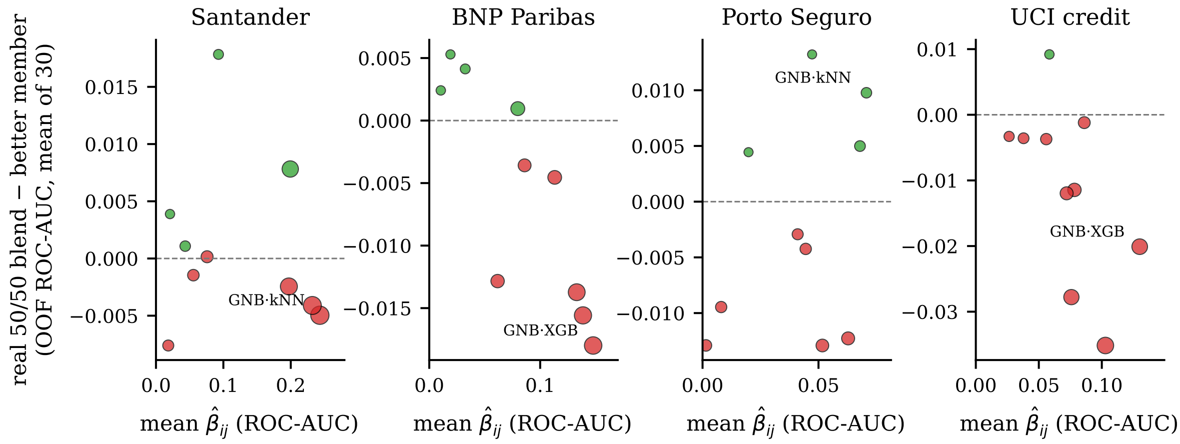

Figure 4: Fitted interaction coefficient against the real gain of the corresponding 50/50 blend over its better member, one point per classifier pair, averaged over 30 partitions. Marker size is the vertex-quality gap of the pair. Green markers denote pairs whose blend beats its better member in at least 80% of partitions. The largest-β pair is labelled; on three of four datasets it lies well below zero.

Table 4: Primary paired comparisons under the weighted cost (30 partitions per dataset; $\Delta > 0$ favours the second-named set: $\Delta\mathrm{IGD}^+ = \mathrm{IGD}^+_{\mathrm{ref}} - \mathrm{IGD}^+_{\mathrm{new}}$, $\Delta\mathrm{HV} = \mathrm{HV}_{\mathrm{new}} - \mathrm{HV}_{\mathrm{ref}}$). Mean with percentile-bootstrap 95% interval, median, wins/ties/losses, matched-pairs rank-biserial r, and the Holm-corrected Nadeau–Bengio p ($\rho = 0.25$, family of eight tests per dataset). Source: `statistics/paired_primary_effects.csv`, `paired_primary_tests.csv`.

		$\Delta\mathrm{IGD}^+$					Δ hypervolume ratio				
Dataset	Comparison	mean [95% CI]	median	W/T/L	r_{rb}	$p_{\mathrm{NB,Holm}}$	mean [95% CI]	median	W/T/L	r_{rb}	$p_{\mathrm{NB,Holm}}$
Santander	NBI-B vs NBI-A	+0.120 [+0.098, +0.143]	+0.125	30/0/0	+1.00	0.007	+0.179 [+0.148, +0.210]	+0.196	30/0/0	+1.00	0.004
	NBI-C vs NBI-B	+0.0016 [+0.0011, +0.0021]	+0.0017	28/0/2	+0.84	0.099	+0.0077 [+0.0064, +0.0089]	+0.0084	28/0/2	+0.98	0.002
	NBI-C vs scalarization	-0.0029 [-0.0050, -0.0010]	-0.0018	10/0/20	-0.47	0.345	+0.0106 [+0.0077, +0.0134]	+0.0104	25/0/5	+0.93	0.062
	NBI-C vs random Dirichlet	+0.820 [+0.624, +1.035]	+0.710	30/0/0	+1.00	0.052	+0.710 [+0.609, +0.807]	+0.778	30/0/0	+1.00	<0.001
BNP Paribas	NBI-B vs NBI-A	+0.239 [+0.111, +0.377]	+0.015	30/0/0	+1.00	0.661	+0.279 [+0.154, +0.415]	+0.058	30/0/0	+1.00	0.661
	NBI-C vs NBI-B	-0.0029 [-0.0044, -0.0011]	-0.0028	6/0/24	-0.74	0.661	+0.0086 [+0.0043, +0.0132]	+0.0106	24/0/6	+0.68	0.661
	NBI-C vs scalarization	+0.397 [+0.293, +0.527]	+0.293	30/0/0	+1.00	0.177	+0.539 [+0.473, +0.606]	+0.510	30/0/0	+1.00	<0.001
	NBI-C vs random Dirichlet	+1.513 [+1.290, +1.744]	+1.512	30/0/0	+1.00	<0.001	+0.943 [+0.912, +0.968]	+0.981	30/0/0	+1.00	<0.001
Porto Seguro	NBI-B vs NBI-A	+0.0380 [+0.0172, +0.0584]	+0.0296	23/0/7	+0.65	0.786	+0.085 [+0.037, +0.131]	+0.076	24/0/6	+0.66	0.786
	NBI-C vs NBI-B	+0.0420 [+0.0243, +0.0630]	+0.0244	28/0/2	+0.90	0.786	+0.108 [+0.064, +0.162]	+0.074	28/0/2	+0.91	0.786
	NBI-C vs scalarization	+1.641 [+0.079, +4.691]	+0.091	30/0/0	+1.00	0.786	+0.270 [+0.188, +0.369]	+0.217	30/0/0	+1.00	0.349
	NBI-C vs random Dirichlet	+1.943 [+1.562, +2.357]	+1.789	30/0/0	+1.00	0.023	+0.930 [+0.886, +0.967]	+0.980	30/0/0	+1.00	<0.001
UCI credit	NBI-B vs NBI-A	+0.0362 [+0.0208, +0.0502]	+0.0499	24/0/6	+0.75	0.452	+0.060 [+0.034, +0.082]	+0.092	24/1/5	+0.75	0.452
	NBI-C vs NBI-B	+0.158 [+0.129, +0.190]	+0.145	29/0/1	+1.00	0.010	+0.277 [+0.226, +0.332]	+0.275	30/0/0	+1.00	0.010
	NBI-C vs scalarization	+0.0008 [-0.0012, +0.0030]	+0.0005	18/0/12	+0.12	1.000	+0.0015 [-0.0031, +0.0067]	+0.0007	16/0/14	+0.06	1.000
	NBI-C vs random Dirichlet	+0.140 [+0.126, +0.155]	+0.140	30/0/0	+1.00	<0.001	+0.252 [+0.232, +0.273]	+0.241	30/0/0	+1.00	<0.001

is convex in $p_{\mathbf{w}}$, so Jensen's inequality forces blending to look favourable. Only the ROC-AUC interaction signs are interpretable, and those are the ones the blend test contradicts.

5.3 RQ3 and RQ4: Surrogate anchors, and whether real anchors repair the front

The primary comparisons are in Table 4; Figure 5 shows the paired distributions and Figure 6 two representative fronts.

Replacing the surrogate-derived anchors with anchors computed from the real single-objective optima, changing nothing else, improves both primary endpoints in every one of the 30 partitions on Santander ($\Delta\mathrm{HV} = +0.179$, 95% CI $[0.148, 0.210]$; Holm-corrected $p = 0.004$) and on BNP Paribas (ΔHV median +0.059), and in 24/30 partitions on Porto Seguro (+0.085, $[0.037, 0.131]$) and UCI credit (+0.060, $[0.034, 0.083]$). The rank-biserial effect size is 1.00 on the two datasets with unanimous wins and 0.65–0.75 on the other two, and the Wilson interval for a 30/30 win fraction is $[0.89, 1.00]$.

Three qualifications keep this from being a stronger claim than the data support. First, the corrected test reaches significance only on Santander (and, for NBI-C versus NBI-B, on UCI

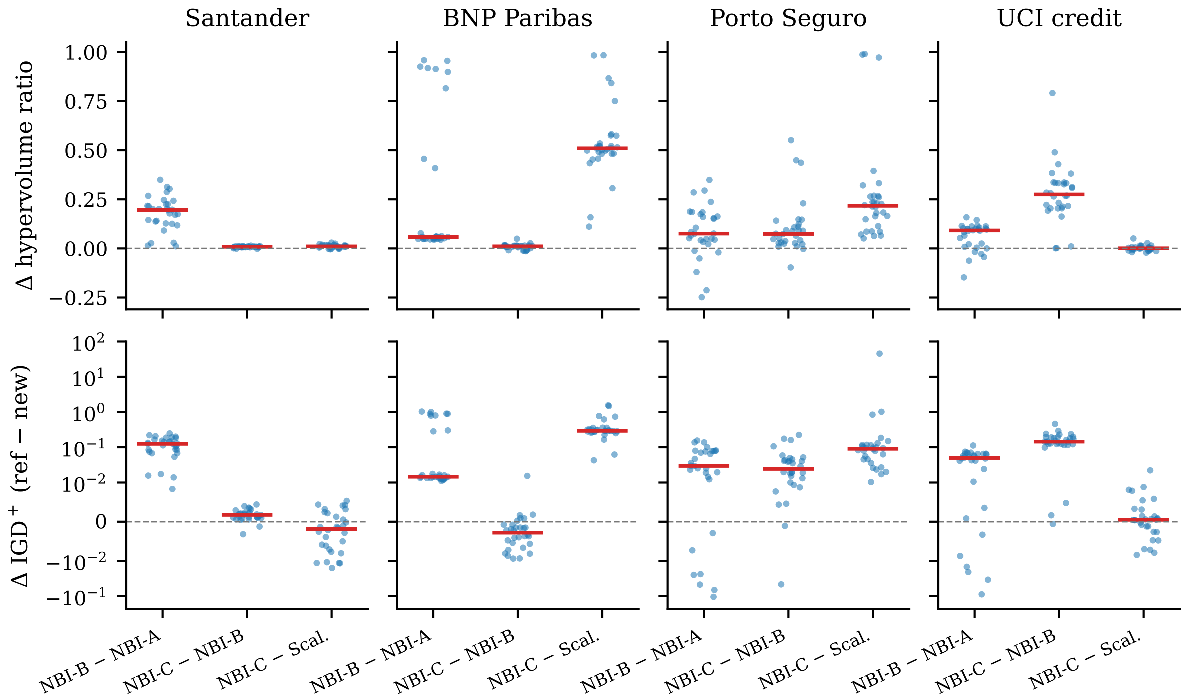

Figure 5: Paired differences on the two primary endpoints, one point per outer partition, red bar at the median. The IGD$^+$ axis is symmetric-logarithmic. The BNP Paribas NBI-B − NBI-A distribution is visibly bimodal: a body of small positive differences and a tail of seven partitions where NBI-A collapses.

credit). On BNP Paribas the effect is unanimous in sign yet the corrected p is 0.66, because the differences are bimodal — the mean +0.279 is over four times the median +0.059 — so by the pre-registered rule we report the median with its bootstrap interval $[0.051, 0.070]$, the win fraction and the rank-biserial correlation instead. Second, the Porto Seguro and UCI credit win rates are driven by the twenty partitions added in the extension (18/20 and 19/20, against 6/10 and 5/10 with one tie in the first ten; on UCI credit the batch difference is itself significant under an identical protocol, Fisher $p = 0.009$). The anchor effect there is real in direction, but its magnitude varies enough between draws of ten partitions that ten were not enough to see it. Third, the effect depends on the cost definition: under the support cost Santander survives unchanged (30/30) and Porto Seguro stays positive on IGD$^+$ (29/30), but BNP Paribas weakens to 19–20/30 and UCI credit *reverses*, NBI-B being worse in 26–27/30 partitions. The full support-cost repetition of Table 4 is supplementary Table S1.

What goes wrong when anchors come from the surrogate. The BNP Paribas bimodality has a specific mechanism. NBI-A's hypervolume ratio splits into 21 partitions at 0.89–0.93, two intermediate, and seven collapsing to 0.04–0.12. The seven coincide exactly with the partitions in which the parsimony rule selects the *quadratic* ROC-AUC order (7/7, against 0/22 of the linear-selected ones). A quadratic surface moves the surrogate AUC anchor off the gradient-boosting vertex to an interior mixture that the empirical reference dominates; in six of the seven that mixture carries k-nearest neighbours and costs 16–31 ms per 1,000 rows, so anchor cost tracks the collapse (Spearman −0.69), but one partition collapses with a cheap anchor, making cost a marker rather than the mechanism.

The consequence for the gate is uncomfortable. Because the quadratic order is selected precisely when it predicts unseen points better, the collapsed partitions have *higher* external R^2_{ext} (median 0.46 against 0.08) and pass the gate *more* often (3 of 7, against 2 of 23) than the partitions where NBI-A works. A gate on surface quality does not protect against this failure;

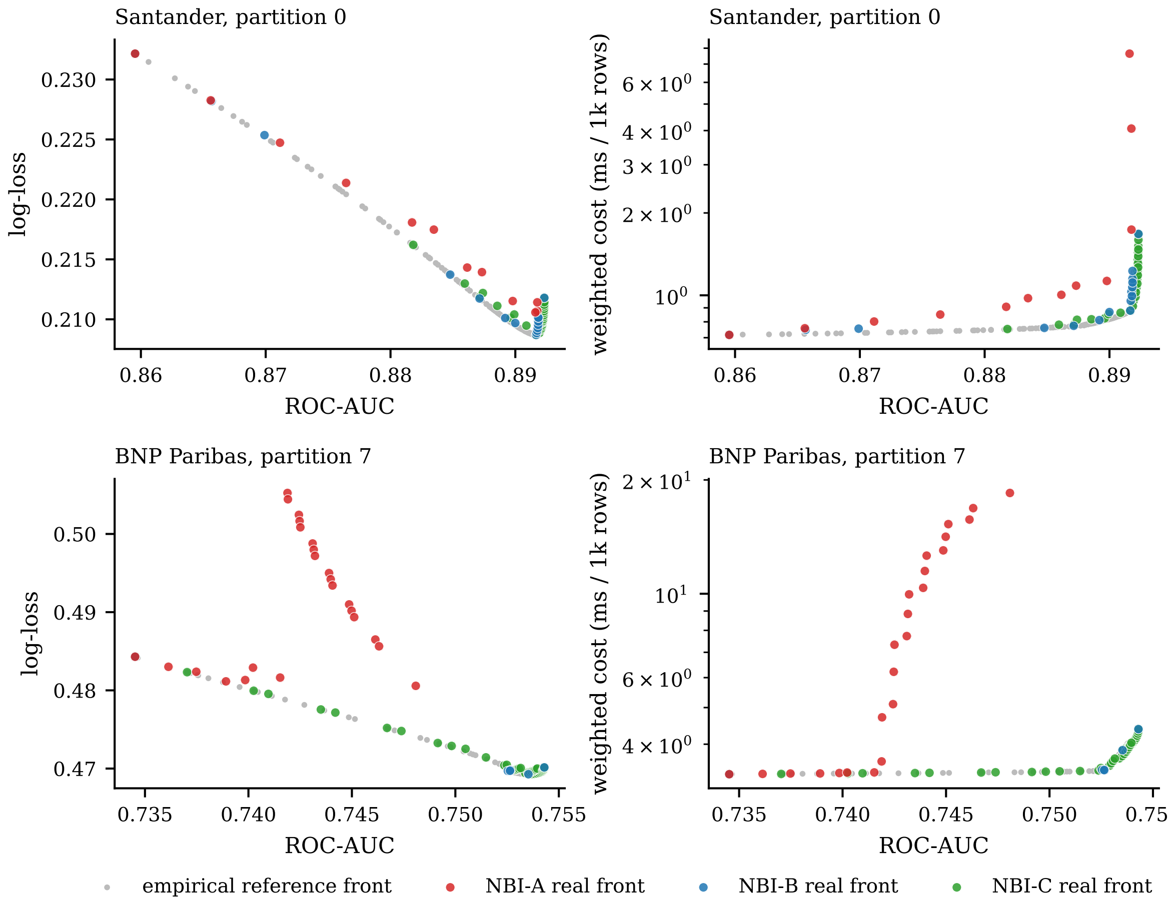

Figure 6: Revalidated fronts of the three NBI variants against the empirical Pareto reference, in the accuracy–calibration and accuracy–cost projections, for one partition of Santander (top) and one BNP Paribas partition in the collapse regime (bottom). Surrogate anchors displace the NBI-A front away from the reference and up the cost axis.

real anchors do, NBI-B reaching 0.935–0.996 in all thirty partitions. This also settles how the gate-conditional analysis must be read: the apparent finding that the anchor benefit is larger where the gate passes on BNP Paribas (+0.64 against +0.21) is NBI-A's collapse restated, since three of the five gate-passing partitions are collapse partitions, and it is not evidence that real anchors add more when the surface is reliable.

5.4 RQ5: Surrogate NBI against metamodel-free NBI

Removing the metamodel and running the identical NBI on the exact out-of-fold objectives gives the best or tied-best approximation on every dataset. NBI-C attains a median hypervolume ratio of 0.989, 0.983, 0.982 and 0.976 on Santander, BNP Paribas, Porto Seguro and UCI credit. It exceeds 0.97 in every partition of Santander and BNP Paribas but in 24/30 and 25/30 partitions of Porto Seguro and UCI credit; the weaker but uniform statement is that it exceeds 0.95 in all thirty partitions of three datasets and in 26/30 on Porto Seguro, whose worst partition reaches 0.877. Against NBI-B it wins on hypervolume in 28, 24, 28 and 30 of 30 partitions; the corrected test reaches significance on Santander ($p = 0.003$) and UCI credit ($p = 0.010$).

Two caveats bound this. On BNP Paribas the arms genuinely split: NBI-C covers more volume (24/30) while NBI-B is closer to the reference in IGD$^+$ (24/30, $\Delta = -0.003$). And the large UCI credit gap is not a surrogate-accuracy effect at all, since UCI credit's surfaces pass the gate in every partition. It is a convergence effect: NBI-B certifies a median of 29 of its 66

Table 5: Mean wall-clock seconds per partition of the main stages (Apple M4 Max, 8 worker threads), the NBI-C/NBI-B time ratio, the mean number of real out-of-fold objective evaluations consumed by NBI-C (NBI-A and NBI-B consume none: they optimize the surrogates), and the mean fraction of the 66 NBI subproblems that terminated successfully (A/B: SLSQP-certified; C: equality-feasible under the lenient rule). Source: `tables/stage_times.csv`, `tables/nbi_runs.csv`.

Dataset	OOF fits	Single-obj. refs	Reference	NBI-A	NBI-B	NBI-C	C/B	Real evals (C)	Success A / B / C
Santander	335	444	243	34	35	2528	73×	412k	0.98 / 0.95 / 0.83
BNP Paribas	86	243	131	23	36	1410	40×	417k	0.95 / 1.00 / 0.83
Porto Seguro	69	441	273	85	56	2505	45×	417k	0.58 / 1.00 / 0.98
UCI credit	11	54	39	125	100	307	3×	426k	0.39 / 0.48 / 0.98

subproblems, indicators use converged candidates only, and the gap correlates with that count at Spearman −0.92. Retaining NBI-B's unconverged iterates preserves the direction (30/30) but cuts the mean gap from +0.277 to +0.108, and in the three partitions where NBI-B converged fully the gap is at most 0.01. Neither observation supports a claim that interior surrogate error dominates once anchors are right.

The premium is large and uneven (Table 5): NBI-C consumes about 4.2×10^5 real objective evaluations per replication where the surrogate arms consume none, at 73, 40, 45 and 3 times the wall-clock time of NBI-B. On UCI credit the premium is small and buys the largest gain; on Santander it is 77-fold and buys +0.008 hypervolume on a dataset whose log-loss surface is already exact. Metamodel-free NBI is the safe default, justified where the surfaces or the anchors fail, not uniformly preferable.

Cheap comparators. Random weighted scalarization on the same surfaces with the same real anchors is strong where the surfaces are reliable: it ties NBI-C on UCI credit (differences within ±0.002, win fractions 16/30 and 18/30, corrected $p = 1.00$), matches it on Santander hypervolume (0.978 against 0.989) and in fact beats NBI-C on Santander IGD$^+$ in 20/30 partitions. It collapses where they are not, falling to a median of one front point on BNP Paribas and a hypervolume ratio of 0.474. This comparison is confounded and we say so: NBI-C optimizes exact objectives while scalarization optimizes surrogates, so its advantage on BNP Paribas and Porto Seguro cannot be attributed to the NBI construction rather than to the objective source. On identical surfaces, NBI-B beats scalarization in 30/30 partitions on BNP Paribas and 25/30 on Porto Seguro but loses 29/30 on UCI credit.

The 66-point random Dirichlet sample loses to every method on every dataset, but this is a floor check and not a statement about search efficiency. Its hypervolume ratio is exactly zero in 10/30, 21/30 and 25/30 partitions on Santander, BNP Paribas and Porto Seguro, because uniform random compositions cost 8 to 20 times more than the reference front. The unoptimized 66-run mixture design also beats it 30/30 everywhere. The comparator is matched to the NBI sets only in the number of candidates returned, and no efficiency claim is made from it.

We also record a negative result. NBI is often motivated by the even spread of the points it generates. After revalidation on the real objectives, that advantage is not present in our data under any spacing definition we computed: the size-matched spacing percentile of NBI-C is 0.99, 0.54, 0.96 and 0.41 across the four datasets, while the random Dirichlet sample scores 1.00, 1.00, 1.00 and 0.93. Even spacing in the CHIM does not survive the map to revalidated objective space.

5.5 RQ6: What is the actual multiobjective conflict?

In objective space the conflict between discrimination and calibration is almost degenerate. The direct ROC-AUC optimum exceeds the SLSQP log-loss optimum by 0.00068, 0.00080, 0.00006 and 0.00034 ROC-AUC, with bootstrap interval half-widths below 0.00004 — 5 to 100 times

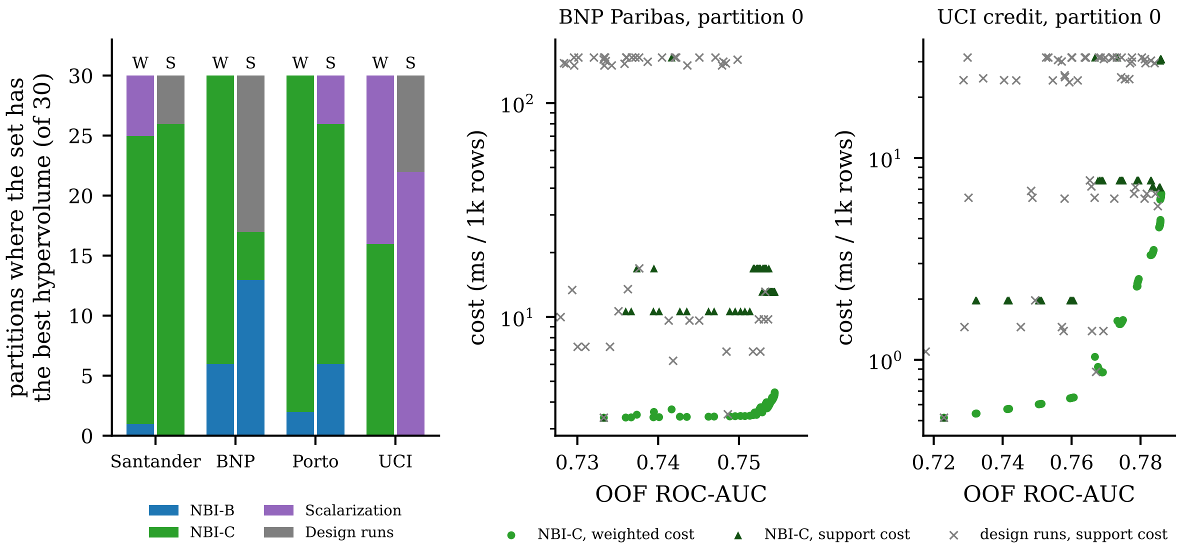

Figure 7: Left: which method attains the best hypervolume, per dataset, under the weighted cost (W) and the support cost (S), counted over 30 partitions. Right: the same NBI-C candidate set scored under both cost definitions against the raw design runs, for one partition each of BNP Paribas and UCI credit. Under the step cost the design runs occupy cheap supports that the NBI sets, optimized on the linear relaxation, do not reach.

smaller than the gain from ensembling at all — and on Porto Seguro the two optima practically coincide (L_1 distance 0.06, support Jaccard 0.97).

Two qualifications prevent the stronger statement that the problems are the same. The optima differ in weight space on the other three datasets (L_1 distance 0.35, 0.38, 0.20; on Santander the AUC optimum adds about 0.12 of random forest to a Gaussian-naive-Bayes-plus-boosting blend), and Santander pays 0.0032 of log-loss for its 0.00068 of ROC-AUC. The practitioner gets a different *model*, even though the objective values are nearly indistinguishable.

We initially read a large deployment-cost difference into this comparison and it did not survive checking. On Santander the AUC optimum appeared to cost +254 ms per 1,000 rows more than the log-loss optimum in 22 of 30 partitions. The cause is an artifact of the activity threshold in Equation (6): the derivative-free AUC search leaves a k-nearest-neighbour weight between 0.0015 and 0.062 that sits just above ε and is AUC-inert — zeroing it changes out-of-fold ROC-AUC by 3×10^{-6} on average, retains at least 93% of the gain and *improves* log-loss. Once such near-threshold weights are cleaned, an AUC-equivalent blend without k-nearest neighbours costs about +4 ms per 1,000 rows. We report the corrected version and flag the artifact, because it is a general hazard: a support-based cost is sensitive to weights that are numerically nonzero and practically irrelevant, and any pipeline using Equation (6) needs an explicit cleaning step.

5.6 RQ7: How does the cost definition affect the front?

Whether a solution is good depends on which cost is being paid, and the two definitions disagree often enough to change conclusions (Figure 7, Table 6).

The hypervolume-best method changes between the two definitions in 10/30 partitions on Santander, 24/30 on BNP Paribas, 8/30 on Porto Seguro and 20/30 on UCI credit. On Santander and Porto Seguro NBI-C remains best under both (26/30 and 20/30). On UCI credit random scalarization takes the support-cost hypervolume in 22/30. Several paired readings reverse outright, as noted in Section 5.3 and Section 5.4.

The mechanism, where it can be isolated, is that a method optimizing $\sum_i w_i c_i$ has no incentive to zero out a component: a small weight on an expensive model is nearly free under the

Table 6: Holdout ranking agreement (partitions in which the OOF-best knee pick among NBI-A/B/C and scalarization is also holdout-best, weighted cost) and cost-definition sensitivity (partitions in which the hypervolume-best set differs between the weighted and the support cost; median Spearman correlation of the set rankings). Wilson 95% intervals. Sources: `statistics/proportion_intervals.csv`, `statistics/cost_definition_sensitivity_r30.csv`.

Dataset	Holdout ranking agreement	Best set differs (W vs S)	Rank ρ (W vs S)
Santander	30/30 [0.89, 1.00]	10/30 [0.19, 0.51]	0.83
BNP Paribas	15/30 [0.33, 0.67]	24/30 [0.63, 0.90]	0.89
Porto Seguro	23/30 [0.59, 0.88]	8/30 [0.14, 0.44]	0.94
UCI credit	30/30 [0.89, 1.00]	20/30 [0.49, 0.81]	0.66

relaxation and fully priced under the step cost. The raw 66-run design, by contrast, contains the pure vertices and the two-model faces by construction, so it occupies exactly the cheap supports the NBI sets miss, and under a step cost it can win outright — as it does in 13/30 BNP Paribas partitions.

One earlier reading of the BNP Paribas support-cost results did not survive scrutiny and we correct it here. The apparent bimodality of NBI-C's support hypervolume is a normalization artifact, not a regime: the per-partition normalization box has a cost bound near 17 ms per 1,000 rows in 13 partitions where no k-nearest-neighbour-bearing point survives on the support reference front, and near 160–167 in the other 17, a tenfold rescaling of one axis. NBI-C's fronts are the same in both groups. Under a common bound the split disappears entirely. The robust statement is qualitative — NBI sets optimized on the linear cost omit the cheapest supports that the raw design contains, and the size of the penalty depends on the chosen cost range — and we make only that claim.

5.7 RQ8: Do out-of-fold-selected solutions transfer to the untouched holdout?

In level, yes. The mean holdout-minus-out-of-fold ROC-AUC of every knee-selected solution lies within ± 0.005 as a point estimate, and the shift equals the weight-averaged shift of the base models on the same partition (correlation 0.97–1.00). It is a partition and refitting effect shared with the constituent models, not selection optimism.

In ranking, it depends on the dataset. The out-of-fold-best of the four knee picks is also holdout-best in 30/30 partitions on Santander, 15/30 on BNP Paribas, 23/30 on Porto Seguro and 30/30 on UCI credit, and the twenty added partitions reproduce this pattern exactly (20/20, 10/20, 15/20, 20/20). The BNP Paribas figure measures how close the four candidates are, not a failure of selection: their median out-of-fold gap is 0.0005 against a holdout noise standard deviation of about 0.0012, several partitions are decided by gaps of 10^{-5} to 10^{-8}, and whenever the two disagree the holdout regret is at most 0.0022 ROC-AUC. These rates are also specific to the selection rule and cost definition — under the support cost they become 28, 28, 17 and 16 of 30 — and should not be read as dataset properties.

5.8 RQ9: Did ten replications already give the right answer?

Tripling the number of replications changed no directional conclusion (Table 7, Figure 8). The $R = 10$ estimate lies inside the $R = 30$ bootstrap interval in 9 of 12 hypervolume cells, 7 of 12 IGD$^+$ cells, all 8 gate frequencies, all 24 leading coefficient means and sign frequencies, all 16 holdout shifts and 11 of 12 NBI success rates; the sign of the median paired effect is identical in all 32 primary cells. The main level shift is on Santander, where IGD$^+$ rose 39–52% in relative terms for every method in partitions 10–29 while hypervolume moved by less than 0.005: the first ten partitions were simply an easier draw, and every method moved together.

Table 7: Robustness from $R = 10$ to $R = 30$: mean hypervolume ratio and IGD$^+$ (weighted cost) of each NBI variant estimated from partitions 0–9 (the frozen $R = 10$ study, tag `pco213-postwork-r10`) and from all 30 partitions, with the $R = 30$ bootstrap interval and whether the $R = 10$ estimate lies inside it. Source: `statistics/r10_vs_r30_stability.csv`.

		Hypervolume ratio			IGD$^+$		
Dataset	Set	$R = 10$	$R = 30$ [95% CI]	inside	$R = 10$	$R = 30$ [95% CI]	inside
Santander	NBI-A	0.846	0.800 [0.767, 0.833]	no	0.0954	0.1321 [0.1078, 0.1573]	no
	NBI-B	0.982	0.980 [0.977, 0.982]	yes	0.0085	0.0121 [0.0101, 0.0142]	no
	NBI-C	0.991	0.987 [0.985, 0.990]	no	0.0069	0.0105 [0.0084, 0.0127]	no
BNP Paribas	NBI-A	0.717	0.695 [0.560, 0.819]	yes	0.2212	0.2502 [0.1234, 0.3877]	yes
	NBI-B	0.966	0.974 [0.969, 0.978]	no	0.0138	0.0114 [0.0096, 0.0132]	no
	NBI-C	0.982	0.982 [0.981, 0.984]	yes	0.0143	0.0142 [0.0134, 0.0151]	yes
Porto Seguro	NBI-A	0.803	0.781 [0.750, 0.812]	yes	0.0751	0.0911 [0.0765, 0.1055]	no
	NBI-B	0.820	0.866 [0.810, 0.913]	yes	0.0695	0.0531 [0.0351, 0.0746]	yes
	NBI-C	0.973	0.974 [0.964, 0.983]	yes	0.0113	0.0111 [0.0075, 0.0155]	yes
UCI credit	NBI-A	0.661	0.639 [0.589, 0.693]	yes	0.1854	0.2043 [0.1724, 0.2340]	yes
	NBI-B	0.670	0.699 [0.643, 0.751]	yes	0.1809	0.1681 [0.1383, 0.2008]	yes
	NBI-C	0.975	0.976 [0.974, 0.979]	yes	0.0107	0.0099 [0.0089, 0.0110]	yes

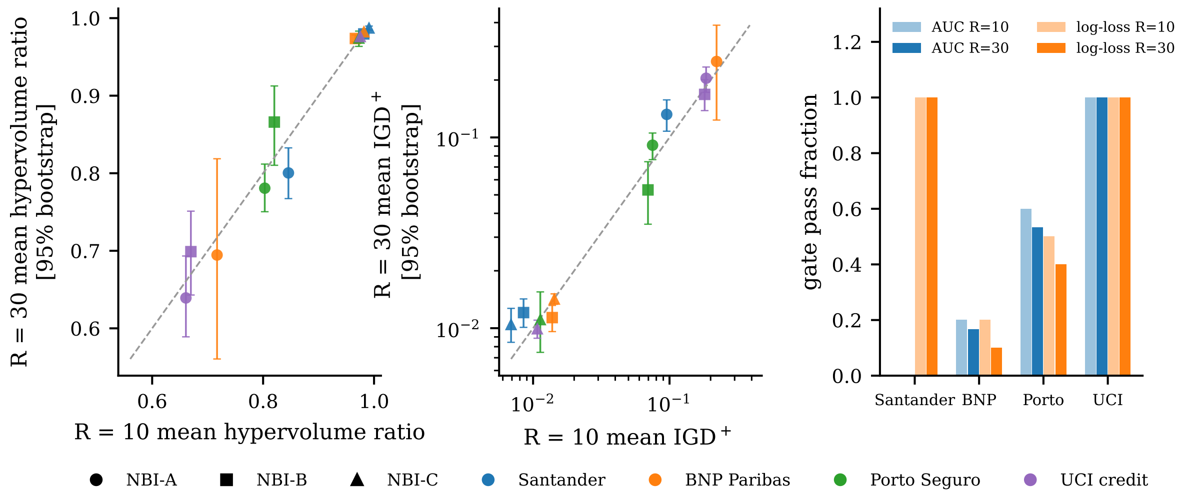

Figure 8: Mean indicator at $R = 10$ against $R = 30$ with bootstrap intervals (left, centre) and gate frequencies at both sample sizes (right). Points on the diagonal are unchanged estimates.

What the extension did deliver is not a change of conclusion but a change of status for four findings. The anchor effect on Porto Seguro and UCI credit went from an undecidable 6/10 and 5/10 to 24/30 with intervals excluding zero. The BNP Paribas anchor-collapse regime went from an observation to a quantified rate (7/30) with an identified trigger (quadratic order selection). The NBI-C versus NBI-B split on BNP Paribas became resolvable as a genuine hypervolume/IGD$^+$ trade-off rather than noise. And two apparent phenomena — the BNP Paribas support-cost bimodality and the Santander deployment-cost gap — were exposed as artifacts.

One caveat on the criterion itself. "The $R = 10$ estimate lies inside the $R = 30$ interval" is lenient, because partitions 0–9 are one third of the $R = 30$ sample. Against the twenty new partitions alone, only 6 of 12 hypervolume and 5 of 12 IGD$^+$ cells agree, and the Porto Seguro and UCI credit anchor win rates differ between batches under an identical protocol. That disagreement is not a defect of either batch; it is the magnitude of partition sensitivity that the replication was designed to measure, and it is the reason we report win fractions and medians rather than means wherever the two diverge.

6 Discussion

The results support a single organizing claim: on the classifier-weight simplex, the mixture-design response surface is a good *description* of ensemble performance and an unreliable *optimization substrate*, and the dominant term in that unreliability is not error in the interior of the surface but error in the points used to build the geometry of the search. This section works through the mechanisms, then states what transfers from the engineering setting where the framework originated and what does not.

6.1 Why anchors dominate

NBI builds its search geometry from the payoff matrix: the utopia point, the objective normalization, the convex hull of individual minima and the quasi-normal direction all derive from the individual optima, and every subproblem in Equation (12) is posed relative to that construction. An anchor error is therefore not local — it relocates the CHIM, redirects every quasi-normal and rescales the objective space in which all 66 subproblems are solved — whereas an error of comparable size in the interior perturbs individual solutions but leaves the frame intact.

That asymmetry predicts what we observe. Replacing surrogate anchors with real ones, holding the surfaces fixed, improves both primary endpoints in 30/30 partitions on the two datasets whose ROC-AUC surfaces almost never pass the gate and in 24/30 on the other two, with hypervolume gains up to +0.18. Removing the surrogate entirely, once anchors are correct, adds much less. The first-order fix is also the cheap one: real anchors are the single-objective optima a practitioner already computes, and on the convex log-loss that is one SLSQP call, so NBI-B costs essentially what NBI-A costs while repairing most of the gap to the metamodel-free arm.

BNP Paribas shows the mechanism clearly. When the parsimony rule selects a quadratic ROC-AUC surface, that surface places its maximum at an interior mixture rather than at the gradient-boosting vertex where the real maximum lies. The anchor moves off the vertex, the CHIM is built between a false utopia point and the true remaining anchors, and the front is dominated almost everywhere by the empirical reference, hypervolume ratio falling from about 0.91 to 0.04–0.12. The failure is geometric rather than numerical — the solver certifies these subproblems — and it is invisible to any diagnostic computed on the surface alone.

6.2 Why the reliability gate does not catch it

The gate asks whether a surface predicts unseen compositions well on average. Anchor placement depends on where the surface puts its *argmax* — a different and much more local property. A surface can be globally accurate and still misplace an extremum, or globally poor with its argmax on the right vertex.

On BNP Paribas the two diverge actively. The parsimony rule selects the quadratic order precisely when it predicts held-out points better, so the collapse partitions have higher external R^2 (median 0.46 against 0.08) and pass the gate more often (3 of 7 against 2 of 23) than the partitions where the surrogate anchor is right: selecting a better-fitting surface makes the optimization worse. This is not a paradox. A linear Scheffé surface has its maximum at a vertex by construction, so the *less* flexible model is accidentally protected, while the more flexible one is free to place an interior maximum the data do not support.

The practical conclusion is a division of labour. An external gate is worth its 100 extra evaluations, cleanly separating unusable surfaces from excellent ones, but it must be paired with an anchor check that is cheaper still: compare each surrogate argmax against the real optimum and prefer the real one. That check is exactly what distinguishes NBI-B from NBI-A, and it removes the failure rather than merely diagnosing it.

6.3 Smooth surrogates against a rank statistic

Two of the three objectives behave as the framework's engineering origins assume: log-loss is convex and smooth in $\mathbf{w}$, and the Brier score is exactly quadratic in $\mathbf{w}$, reproduced with $R^2_{\text{ext}} = 1.000$ in all 120 replications. The ROC-AUC is not. It is a rank statistic, constant wherever the induced ordering does not change, with kinks at the boundaries between orderings, and a low-order polynomial cannot represent a piecewise-constant function with many small steps — it smooths across them. That is why the Santander ROC-AUC surface achieves a Spearman correlation of 0.830 while its external R^2 is negative: it gets the ordering broadly right and the values wrong.

The consequences propagate. Section 5.2 shows a systematic, one-directional bias: on 20 of 40 dataset-pair cells the surface predicts an edge blend superior to both pure components where the real objective delivers none, and in no case does it miss a real one. A quadratic fitted over an edge with one weak endpoint absorbs that curvature into β_{ij} and overshoots its maximum — the same over-prediction of interior optima that displaces surrogate anchors off vertices.

This is the concrete sense in which the transfer is non-trivial. In the originating application the responses are physical measurements, genuinely smooth in the process factors, for which a quadratic is a defensible local model. Discrimination metrics are not, and no experimental design repairs a misspecified model class. Where the objective *is* smooth in the weights, the inherited machinery works as advertised: Santander's log-loss surface passes the gate in every partition.

6.4 Non-smoothness also constrains the optimizer

The same property that defeats the surrogate constrains the metamodel-free arm. SLSQP on a piecewise-constant objective cannot certify optimality, because the gradient is zero almost everywhere. NBI-C therefore accepts a feasible iterate whose equality residual is below 10^{-3}, which is a weaker guarantee than the certification NBI-A and NBI-B obtain on their smooth polynomials. The reported success rates are not on a common scale across arms, and we have been careful not to read the difference as evidence about surrogate quality.

The UCI credit result is the sharpest illustration of why that care is needed. It looks like the strongest evidence in the paper for metamodel-free NBI: a +0.277 hypervolume gain over NBI-B with a Holm-corrected p of 0.010. But UCI credit is the dataset whose surfaces pass the gate in every partition, so surrogate error cannot be the explanation. It is not: NBI-B certifies a median of 29 of its 66 subproblems there, the gap correlates with the number of converged subproblems at Spearman −0.92, and in the three partitions where NBI-B converges fully the gap disappears. A study that reported only the headline number would have drawn exactly the wrong mechanistic conclusion.

6.5 Compute against fidelity

Metamodel-free NBI reaches median hypervolume ratios of 0.976–0.989 and is best or tied-best on the primary endpoints, so it is a defensible default. But it consumes about 4.2×10^5 real evaluations per replication against none for the surrogate arms, at 3 to 73 times the wall-clock cost, and that spend is strongly conditional: on UCI credit the threefold premium buys the largest gain, while on Santander a seventy-three-fold premium buys +0.008 hypervolume on a dataset whose log-loss surface is already exact.

Its affordability is specific to this problem and should not be generalized. Because the ensemble is linear in $\mathbf{w}$ and the base-model probabilities are cached, one evaluation is a matrix–vector product plus a sort — 0.37 to 2.13 ms here. Where each evaluation requires refitting a model the ordering reverses and the surrogate is the only viable route, which is the regime the framework was designed for. The recommendation is therefore conditional on evaluation cost: when real evaluations are cheap, use them, at least for the anchors; when they are expensive, use a surrogate but validate it externally and anchor it on whatever real information is affordable, because anchors are where surrogate error hurts most and they are the smallest number of real evaluations one can spend.

6.6 Response surfaces as descriptions rather than optimizers

The surfaces are strikingly reproducible. Over 30 resampled partitions the sign of each of the four leading interaction coefficients is constant in all eight dataset–response cells, the leading term is identical at $R = 10$ and $R = 30$ in all eight, and coefficient means move by at most 6%. As a description of how ensemble performance varies over the simplex, and in particular of which vertices are weak and which regions are flat, the model is stable and interpretable in a way that a black-box optimizer is not. That interpretability is a genuine benefit of the mixture-design formalism and it survives everything else in this paper.

What does not survive is the step from description to inference about blends. The classical synergism criterion, $\beta_{ij} > |\beta_i - \beta_j|$, is correct as stated and correctly guards against reading the sign of β_{ij} alone; but it is computed *from the surface*, and on these ensembles it inherits the surface's systematic over-prediction of interior optima. Confirming a candidate pair on the real objective costs one matrix product, and that confirmation should be treated as part of the interpretation rather than as an optional check.

6.7 Deployment cost is not the cost that was optimized

Every method here minimized the continuous relaxation $\sum_i w_i c_i$, because a step cost is not differentiable and cannot enter a gradient-based multiobjective formulation. Deployment pays the step cost. The two are not monotonically related on the interior of the simplex, and the consequence is not cosmetic: the hypervolume-best method changes between the two definitions in 10, 24, 8 and 20 of 30 partitions, and several paired conclusions reverse, including the anchor effect on UCI credit.

The mechanism is that a linear cost gives no incentive to zero a component. A weight of 0.01 on a model costing 200 ms adds 2 ms under the relaxation and 200 ms in deployment. Methods that optimize the relaxation therefore drift into the interior and systematically miss the cheap low-support solutions — the pure vertices and two-model faces — that the unoptimized 66-run design contains by construction, which is why that design wins the support-cost hypervolume outright in 13 of 30 BNP Paribas partitions.

The related literature prices deployment either as a support cost over a fixed selected set or as a per-example cascade, and the sparsity-inducing formulations that do use a continuous relaxation of a support count generally do not audit whether the relaxation and the support agree. Where that question has been examined in a related setting, the answer went the other way: for budgeted learning with feature re-use, a support-cost programme can be shown to be solved exactly by its linear relaxation (Nan et al., 2015). Our setting does not inherit that guarantee, and the practical recommendations are concrete. Report which cost model is used; add a cleaning step that zeroes weights below the activity threshold and re-evaluates, since an inert weight just above ε produced a spurious 250 ms difference in our own analysis before we caught it; and if deployment cost is the real objective, consider optimizing over supports explicitly rather than hoping the relaxation lands on a cheap one.

6.8 What the framework's transfer teaches

The pipeline transfers cleanly where its assumptions hold and fails where they do not, and the assumptions are identifiable in advance rather than only in hindsight.

It transfers when the response is smooth in the mixture components. Log-loss and Brier behave exactly as the engineering responses do, and on the datasets where the ROC-AUC surface is also adequate — UCI credit, where it passes in 30/30 — surrogate NBI with real anchors is within 0.03 hypervolume of the metamodel-free arm and random scalarization on the same surfaces is statistically indistinguishable from it. In that regime the inherited machinery is not merely usable, it is the efficient choice.

It does not transfer unexamined when the response is a rank statistic, when the decision variables are the mixture itself so that anchors are vertices of the feasible region rather than interior points, or when an objective is a step function of the support. These are not exotic special cases; they are generic properties of machine-learning objectives, and each of them breaks a different part of the pipeline: model class, geometry, and objective definition respectively.

What we would carry forward into any future application of this framework is not a method but a checklist: validate the surface on compositions it has not seen and record the result; compare each surrogate argmax against the real optimum and prefer the real one; revalidate every candidate on the true objective before scoring; build a reference that no method under test can influence; and state which cost you are paying. None of these is expensive relative to the pipeline they protect, and in this study each of them changed a conclusion that would otherwise have been reported.

7 Limitations

Four datasets is the binding constraint. The statistical unit is one outer partition, but the unit of *generalization* is the dataset, and there are four. Every replication-level interval here describes partition sensitivity within a dataset, not uncertainty about a population of datasets, so every cross-dataset mechanism statement — most importantly "real anchors matter most where the surrogate is unreliable" — is an association over four points rather than a tested hypothesis. All four are binary tabular problems of 3×10^4 to 2×10^5 rows, three from Kaggle credit- and insurance-risk competitions. Nothing here establishes behaviour on multiclass problems, on text or image data, on very small samples, or on ensembles of more than five members, where the simplex dimension and the design size grow quickly.

The model zoo is fixed and shapes the coefficient findings. Base learners and hyperparameters were fixed in advance with no search, so the study measures the weighting layer with the model layer held constant. The zoo deliberately contains a poorly calibrated member and an expensive one, which is what makes the cost objective non-trivial and also what drives the β_{ij} result of Section 5.2, since the largest interactions attach to the weakest vertex. A more homogeneous zoo would produce smaller interaction coefficients and less for the vertex-gap explanation to explain. That finding is therefore conditional on component heterogeneity, and these data cannot say how it behaves for a zoo of near-equal members.

The Pareto reference is empirical. With no analytical front available, all convergence and coverage indicators are computed against a sample-based reference of about 10^5 points per replication with a displacement check. Residual displacement is small but non-zero: 116 of 120 replications met the 5% tolerance, four stopped at the three-round cap with at most 6%, and the objective-space effect is bounded at 0.2% of hypervolume. More importantly, the reference is the non-dominated union of the sample with all candidate sets, so methods contributing many points — NBI-B and NBI-C contribute 30–40% — are partly graded against their own output. This inflates their absolute indicator values without affecting the paired contrasts on which every primary claim rests, but absolute statements about hypervolume ratio should be read with it in mind.

The three NBI arms are not compute-matched, and NBI-C is not matched to anything. NBI-A and NBI-B consume no real objective evaluations; NBI-C consumes about

4.2×10^5 per replication and 3 to 77 times the wall-clock time of NBI-B; the two cheap comparators consume about 66 evaluations and a few seconds. The comparison is therefore between *methods as specified*, not between methods at equal budget, and no efficiency claim is made from it. The random Dirichlet comparator in particular is matched only in the number of candidates returned and is degenerate under the weighted cost; it is a floor check on the reference geometry. A properly budget-matched study — for instance, random search or an evolutionary method given 4×10^5 real evaluations — would answer a different and also worthwhile question, and we do not answer it here.

Solver semantics differ between the arms. NBI-A and NBI-B optimize smooth polynomials and their subproblems are counted as successful only when SLSQP certifies optimality. NBI-C optimizes a piecewise-constant ROC-AUC, where such certification is generically unattainable, so a subproblem is accepted when its equality residual falls below 10^{-3}, under a reduced iteration budget. The reported "success rates" are therefore not on a common scale across arms, and the UCI credit gap between NBI-C and NBI-B is substantially a convergence artifact of NBI-B rather than evidence about surrogate quality (Section 5.4). We report both the filtered and unfiltered versions of that comparison for this reason.

The support cost is a threshold approximation. Deployment cost is modelled by Equation (6) with $\varepsilon = 10^{-3}$, chosen before the study and never changed. It is a proxy: real deployment costs include memory, batching, model loading and shared preprocessing, and it treats a weight of 1.1×10^{-3} as a full model and one of 0.9×10^{-3} as free. Section 5.5 shows this is not hypothetical — an AUC-inert k-nearest-neighbour weight just above the threshold produced a spurious 250 ms cost difference. Any use of a support cost needs a weight-cleaning step, and conclusions that depend on it are sensitive to ε and to the cost range that sets the normalization box.

Cost measurements are single-machine and single-implementation. All c_i are wall-clock medians on one workstation with a fixed library stack and batch size, and would change with hardware, batch size, a compiled inference runtime, or an approximate-nearest-neighbour index replacing brute-force search — which would remove the largest cost asymmetry in the study. The cost objective is realistic for the setting measured, not universal.

No claim of universal dominance, and confounded comparators. NBI-C is best or tied-best on the primary endpoints under the weighted cost on these four datasets; it ties with random scalarization on UCI credit, loses to it on Santander IGD$^+$, and loses its support-cost advantage on BNP Paribas and UCI credit. Its advantage over scalarization on BNP Paribas and Porto Seguro is confounded with the objective source, since NBI-C optimizes exact objectives and scalarization optimizes surrogates; disentangling the NBI construction from the objective source would require a scalarization arm run on the cached out-of-fold objectives, which we did not run.

Selection rules and the holdout analysis. Holdout transfer is reported for a knee rule under the weighted cost, with TOPSIS and the support cost as sensitivity analyses, and the agreement rates differ noticeably between these configurations. They characterize how close the candidates are relative to holdout noise, not the general reliability of out-of-fold selection.

No evolutionary multiobjective baseline. The comparators are the ones needed to decompose the pipeline (surrogate versus real objectives, surrogate versus real anchors, scalarization on the same surfaces, random search as a floor). A canonical evolutionary baseline such as NSGA-II or MOEA/D is absent. Our goal is diagnostic rather than competitive, and adding one would not change any claim made here, but its absence limits what can be said about how this pipeline compares with the standard multiobjective toolbox. We assess the case for adding one in the supplementary baseline-gap assessment.

8 Conclusion

We transferred an established DoE–RSM–NBI framework — mixture design over combination weights, Scheffé surrogate of performance over the simplex, Normal Boundary Intersection on that surrogate — from the engineering and forecasting settings in which it was developed to classifier ensemble weighting, and asked not whether it produces a Pareto front but under which conditions the front it produces can be believed. Answering that required decomposing the pipeline into three arms differing only in where the objectives and the anchors come from, revalidating every candidate on the true out-of-fold objectives, scoring against an empirical reference the methods cannot influence, and repeating the whole procedure on 4 datasets × 30 partitions.

The framework is not uniformly safe, and it is not uniformly unsafe. Its behaviour is governed by three identifiable properties of the problem rather than by anything about the optimizer.

Model class. Where the response is smooth in the weights, the surrogate is excellent: log-loss surfaces pass an external reliability gate in every partition on two of four datasets, and the Brier score, which is exactly quadratic in the weights, is reproduced with $R^2_{\text{ext}} = 1.000$ in all 120 replications. Where the response is a rank statistic, it can fail completely: the Santander ROC-AUC surface never passes the gate and has negative external R^2 despite a rank correlation of 0.83. Surrogate adequacy is a property of the dataset *and* the metric, must be tested on compositions the surface has not seen, and cannot be assumed from in-sample fit.

Geometry. Anchor construction, not interior surrogate accuracy, is the first-order failure mode. Replacing surrogate anchors with real single-objective optima — changing nothing else — improved both primary endpoints in every one of 30 partitions on the two datasets whose surfaces fail, and in 24 of 30 on the two where they largely succeed. Removing the metamodel altogether, once the anchors are right, adds far less. Because NBI derives its utopia point, its normalization and its search directions from the payoff matrix, an anchor error relocates the entire construction, while an interior error perturbs individual subproblems. The practical consequence is favourable: the first-order fix is also the cheapest, since real anchors are the single-objective optima a practitioner already computes.

Diagnosis. An external gate on surface quality is worth its 100 extra evaluations and cleanly separates usable from unusable surfaces, but it does not predict anchor failure, and on one dataset the two properties actively diverge: the partitions in which surrogate-anchored NBI collapses are exactly those in which the parsimony rule selects a surface that fits held-out compositions better and passes the gate more often. Surface accuracy and argmax accuracy are different properties, and only the second governs NBI. The gate should therefore be paired with an anchor check that compares each surrogate argmax against the real optimum — which is simply the difference between our two surrogate arms.

Two further results are practical rather than methodological. The classical synergism criterion $\beta_{ij} > |\beta_i - \beta_j|$ is satisfied by the fitted surfaces for the leading classifier pair on all four datasets in every partition, yet the real 50/50 blend beats its better member on only one of them; across all pairs the surface over-predicts blending gains twenty times and under-predicts none. A large interaction coefficient is a statement about the fitted surface, and on these ensembles the surface systematically overstates what blending buys, so a candidate pair must be confirmed on the real objective before it is called complementary. Separately, the continuous cost that a differentiable method can optimize and the step cost that deployment pays disagree about the best method in 10, 24, 8 and 20 of 30 partitions across the four datasets, and several conclusions reverse between them; any claim about ensemble cost must name which cost it means, and a weight-cleaning step is necessary because an inert weight just above the activity threshold produced a spurious 250 ms difference in our own analysis before we caught it.

Tripling the replications from 10 to 30 changed no directional conclusion. It converted three undecided findings into quantified ones, exposed two apparent phenomena as artifacts, and supplied intervals; it did not alter what the study says. We report that explicitly because it is evidence about how much replication such a study needs, and the answer appears to be that ten partitions identify the conclusions while thirty are required to defend their magnitudes.

What we would carry into any future use of this framework is not a method but a set of controls: validate the surface on compositions it has not seen; compare each surrogate argmax against the real optimum and prefer the real one; revalidate every candidate on the true objective before scoring; build a reference that no method under test can influence; and state which cost is being paid. Each is inexpensive relative to the pipeline it protects, and in this study each of them changed a conclusion that would otherwise have been reported. The framework remains a good *description* of the ensemble-weight simplex — its coefficients are stable to a few percent across thirty resamplings and they say something interpretable about which components are weak. Whether it is also a good *optimizer* there depends on conditions that can now be stated in advance and checked cheaply.

Reproducibility statement

Code and frozen states. All code, manifests, aggregated tables, statistics and figures are in the repository `caioribeiro99/doe_nbi_hpo_project`. Two annotated tags freeze the study. The tag `pco213postworkr10` marks the completed ten-replication analysis; the tag `pco213-postworkr30` (commit `b3ed050`) marks the thirty-replication analysis reported here, and every number in this paper is computed from artifacts at that commit. The manuscript, its figures and its tables are on the branch `paper/surrogatenbiensembleweighting`, cut from that tag. The comparison of the two tags is the content of Section 5.8.

Pipeline and resumption. The benchmark is driven by a single checkpointed runner (`scripts/pco213_run_postwork_benchmark.py`) with ten stages per replication: out-of-fold estimation, mixture design, Scheffé fitting, single-objective references, empirical Pareto reference, NBI-A, NBI-B, NBI-C, comparators and quality scoring. State is recorded per dataset × replication × stage, so an interrupted run resumes without recomputation and a completed stage is never silently recomputed. A dry-run mode lists pending work; it was used to verify, before the extension was launched, that replications 0–9 were complete and that exactly 80 replications (800 stages) were pending. Aggregation and statistics are separate scripts (`pco213_postwork_benchmark_report.py` and `..._stats.py`), so the analysis can be rerun without re-executing the experiment. The library has 68 unit tests, including reference implementations checked against scikit-learn, exact hypervolume on known cases, the 66-run design, NBI anchor and warm-start behaviour, and the resumption planner.

Determinism. Every stochastic component is seeded as $20260904 + r$ for replication r: the outer stratified partition, the inner fold assignment, the model random states, the Dirichlet validation compositions (offset +1000 and +2000), the direct AUC search (+3000, +4000), the empirical reference (+5000 onwards) and the comparators (+12000). The seed policy and the full seed map are stored in the master manifest, together with the configuration history and the resolved library versions.

Environment. Python 3.11.15 with NumPy 2.4.6, pandas 3.0.5, SciPy 1.17.1, scikit-learn 1.9.0, statsmodels 0.15.0 and XGBoost 3.2.0, on macOS (Apple M4 Max, arm64), with eight worker threads. The complete execution consumed 81.1 hours of stage time (27.5 for the first ten replications, 53.6 for the extension) and produced 3,600 model fits, 19,920 design evaluations, 23,760 NBI subproblems, 5.02×10^7 real objective evaluations, 1.55×10^7 reference points and 4.8×10^6 direct-AUC-search evaluations. Two stages were retried during the first ten replications and none during the extension.

Data availability and redistribution limits. The raw datasets are not redistributed. The UCI default-of-credit-card-clients data are downloadable from the UCI Machine Learning Repository. The three Kaggle datasets are subject to competition rules and must be obtained from Kaggle by the user; the repository records, for each, the exact source, the acquisition command, the expected row and feature counts, the expected positive-class prevalence and the SHA-256 checksum of the file actually used, so an independent copy can be verified byte-for-byte before use. Where a competition endpoint was unavailable, the file was obtained from a public mirror dataset and validated against those invariants; the manifests record which source was used for each dataset. Competition test labels are never used. Porto Seguro was subsampled to 200,000 rows by a stratified draw with the recorded seed. The per-replication raw artifacts (out-of-fold probability matrices, candidate sets, reference samples) total about 829 MB and are deliberately not version-controlled; they are fully reconstructible from the seeds and the runner.

What a replication requires. Reproducing the aggregate results from the published tables and statistics requires only the repository. Reproducing them from raw data requires the four datasets, the pinned environment and approximately 81 hours of single-workstation compute, or proportionally less for a subset of datasets or replications, since the runner is resumable and parameterized by dataset and replication count.

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